

Optimally Imprecise Memory and Biased Forecasts[†]

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We propose a model of optimal decision-making subject to a memory constraint in the spirit of models of rational inattention. Our theory differs from that of Sims (2003) in not assuming costless memory of past cognitive states. The model implies that both forecasts and actions will exhibit idiosyncratic random variation; that average beliefs will exhibit a bias that fluctuates forever; and that more recent news will be given disproportionate weight in forecasts. The model provides a simple explanation for the overreaction to news observed in the laboratory by Afrouzi et al. (2023). (JEL D83, D84, D91, G12, G41)

The hypothesis of rational expectations (RE) proposes that decisions are based on expectations that make use of all available information in an optimal way: that is, those that would be derived by correct Bayesian inference from an objectively correct prior and the data that has been observed to that date. Yet, both in surveys of individual forecasts of macroeconomic and financial variables and in forecasts elicited in experimental settings, beliefs are more heterogeneous than this hypothesis should allow, and forecast errors are predictable on the basis of variables observable by the forecasters, contrary to this hypothesis. In particular, a number of studies have argued that forecasts typically overreact to new realizations of the variable being forecasted. (See Bordalo et al. 2020 and Afrouzi et al. 2023, for recent examples with extensive references to prior literature.)

A variety of models of expectation formation have been proposed that allow for such overreaction. The simplest type of model simply posits that the forecasted future value of a variable is a particular linear function of the last few observations of the variable; with an appropriate choice of the coefficients (such as those proposed by Metzler 1941), a forecasting heuristic of this kind may imply that a recent increase in the variable will be extrapolated into the future, so that further increases are anticipated, regardless of whether the degree of serial correlation of changes in the variable make this an optimal forecast. A classic critique of such proposals, however, is that of Muth (1961): why should decision-makers continue to forecast in this way, if their forecasts are systematically biased, as repeated observations should

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eventually make clear? Moreover, a mechanical heuristic with fixed coefficients is unable to explain how the biases in observed forecasts change depending on the persistence of the process that is forecasted (Afrouzi et al. 2023).

Fuster, Laibson, and Mendel (2010) and Fuster, Hébert, and Laibson (2011) propose a more sophisticated model, in which decision-makers are assumed to forecast a time series by modeling it as a low-order autoregressive process, that may not capture the complexity of the actual dynamics of the series. This approach, however, does not explain why only a particular subset of lagged values of the series are considered as possible regressors in the forecasting model. Moreover, the Fuster et al. explanation implies that overreaction should only be observed in the case of variables that are not well described by an $AR(k)$ process of low enough order. Yet Afrouzi et al. (2023) find significant overreaction in an experiment in which the true data generating process is an $AR(1)$ process; in fact, they find the most severe degree of overreaction (as discussed further below) when the process to be forecasted is white noise.

Here we offer a different explanation for the pervasiveness of overreaction.¹ We consider a model in which a decision-maker's forecasts (or more generally, actions with consequences that depend on the future realization of some variable) can be based both on currently observable information and an imperfect memory of past observations. Subject to this constraint on the information that the decision rule can use, we assume the rule is optimal. Moreover, rather than making an arbitrary assumption about the kind of statistics about past experience that can be recalled with greater or lesser precision, we allow the memory structure to be specified in a flexible way, and assume that it is optimized for the particular decision problem, subject only to a constraint on the overall complexity of the information that can be stored in (and retrieved from) memory—or more generally, subject to a cost of using a more complex memory structure.

In the limiting case, in which the cost of memory complexity is assumed to be negligible, the predictions of our model coincide with those of the rational expectations hypothesis. But when the cost is larger (or the constraint on memory complexity is tighter), our model predicts that forecasts should be both heterogeneous (even in the case of forecasters who observe identical data) and systematically biased. Moreover, the predicted biases include the type of overreaction to news documented in laboratory forecasting experiments by Afrouzi et al. (2023). And unlike the models of Fuster, Laibson, and Mendel (2010) and Fuster, Hébert, and Laibson (2011), our model predicts that overreaction to news will be most severe in the case of time series exhibiting little serial correlation.

In seeking to endogenize the information content of the noisy cognitive state on the basis of which people must act, our theory is in the spirit of Sims's (2003) theory of "rational inattention"; and indeed, we follow Sims in modeling the complexity constraint using information theory.² There is nonetheless an important difference between our theory and that of Sims (2003). Sims assumes a constraint on the precision with which new observations of the world can reflect any current or past conditions outside

¹ A preliminary version of our analysis, treating only the case of forecasts of a serially uncorrelated process, appeared as Azeredo da Silveira and Woodford (2019).

² Our model of the constraint on the precision of memory as a mutual information constraint is similar to the proposal of Fox and Tishby (2012), though we develop the analysis in a way that is fairly different from theirs.

the head of the decision-maker, but assumes perfectly precise memory of all of the decision-maker's own past cognitive states, and also assumes that past external states can be observed at any time with the same precision as current conditions. We instead assume (for the sake of simplicity) that the current external state can be observed with perfect precision, but that memory of past cognitive states is subject to an information constraint; and we further assume that the decision-maker has no access to external states that occurred in the past, except through (information-constrained) access to her own memory of those past states. These differences are crucial for the ability of our model to explain overreaction to news.

We assume that in each period t , a decision-maker's (DM's) only access to the past is through a retrieved memory state m_t , which provides a noisy record of (aspects of) the DM's cognitive state s_{t-1} in the previous period. The DM's cognitive state s_t in period t , upon which period t actions are based, then consists both of the information in m_t and the current observation of the external state y_t . A noisy record of this cognitive state can then be retrieved in period $t + 1$, and so on. Our model differs from one of optimal Bayesian decision-making only in our assumption that the memory state m_t is not perfectly informative about the prior cognitive state s_{t-1} . The noise in the retrieved memory state is endogenized subject to an assumed cost of memory precision.

The model describes a DM with no access to information about the past, apart from the imprecise summary that can be obtained by accessing their memory. It is most plausibly applicable to the situation of a subject in a laboratory experiment; we present a quantitative application to data of this kind below. But we believe that the model can also be reasonably applied to situations of greater economic consequence, in which people draw conclusions from their personal experience without seeking other sources of information. For example, this would be the way most households form expectations about their future income prospects when making spending decisions.

The model, as presented, is less obviously applicable to the situations of professional forecasters who can access extensive bodies of recorded data. Note that in our model, the currently observable external state is assumed to be accessible with perfect precision. This may be a reasonable approximation (and certainly simplifies calculations) when, as assumed here, the observable external state is a single number; it is less obviously plausible when the external data that can in principle be observed are a high-dimensional vector, as in the situation of the professional forecaster. In order to treat such a case, it is important to allow for limits on the precision of access to currently observable information, along with the memory limits that we emphasize here. Sung (2024) presents such an extension of the model, and shows that the extended model can explain some of the biases observed in professional forecasts.

Other recent papers that explore the economic consequences of assuming that memory allows only a noisy recollection of past observations include Neligh (2024) and Afrouzi et al. (2023). While these authors also assume that some aspects of memory structure are optimized for a particular decision problem, the classes of memory structures that they consider are different than the one that we analyze here, as discussed further in Section VIB.

We present the assumptions of our model and state the optimization problem that we consider in Section I. Section II offers a characterization of the optimal memory structure in our model, starting with a simple environment that yields a

closed-form solution. We emphasize the failure of beliefs ever to converge to those associated with a rational expectations equilibrium. Section III extends the analysis in a more general setup and derives the optimal dimension and the structure of the memory state. Section IV illustrates the model's implications, discussing quantitative aspects of numerical solutions under two different examples of memory costs. Section V presents the quantitative predictions of the model for statistics of the kind reported by Afrouzi et al. (2023), showing not only that the model can produce overreaction to news, but that it can be parameterized so as to predict roughly the degree of overreaction measured by these authors. Finally, Section VI discusses how our model compares to other related models. Section VII concludes.

I. A Flexible Model of Imprecise Memory

Here we introduce the class of linear-quadratic-Gaussian decision problems that we study, and specify the nature of a general constraint on the precision of memory. This gives rise to a dynamic optimization problem, the solution to which we study in the following sections.

A. The Decision-Maker's Problem

In the kind of decision problem which we consider, a decision-maker (DM) observes the successive realizations of a univariate stochastic process y_t ("the external state"), which we assume to be a stationary AR(1) process. We write the law of motion of this process in the form

$$(1.1) \quad y_t = \mu + \rho(y_{t-1} - \mu) + \epsilon_{yt},$$

where μ is the mean, ρ is the coefficient of serial correlation (with $|\rho| < 1$), and $\{\epsilon_{yt}\}$ is an i.i.d. sequence, drawn each period from a Gaussian distribution $N(0, \sigma_\epsilon^2)$. The variance of the external state (conditional on the value of μ and the other parameters) will therefore equal $\sigma_y^2 \equiv \sigma_\epsilon^2 / (1 - \rho^2)$.

The decision-maker produces each period a vector of forecasts z_t , so as to minimize the expected value of a discounted quadratic loss function

$$(1.2) \quad E \sum_{t=0}^{\infty} \beta^t (z_t - \tilde{z}_t)' W (z_t - \tilde{z}_t),$$

where W is a positive definite matrix specifying the relative importance of accuracy of the different dimensions of the vector of forecasts. The eventual outcomes that the DM seeks to forecast (the elements of the vector $\tilde{z}_t$) are each functions of the future evolution of the external state,³

$$\tilde{z}_{i,t} \equiv \sum_{j=0}^{\infty} a_{ij} y_{t+j},$$

³Note, the variables denoted $\tilde{z}_t$ are not quantities the value of which is determined at time t ; the subscript t is used to identify the time at which the DM must produce a forecast of the quantity, not the time at which the outcome will be realized. Thus, the best possible forecast of $\tilde{z}_t$ at time t , even with full information, would be given by $E_t \tilde{z}_t$, which will generally not be the same as the realized value $\tilde{z}_t$.

where the coefficients $\{a_{ij}\}$ satisfy $\sum_j |a_{ij}| < \infty$ for each i , and the discount factor satisfies $0 < \beta < 1$. This formalism allows us to assume that the DM may produce forecasts about the future state at multiple horizons (as is typically true in surveys of forecasters, and also in the experiment of Afrouzi et al. 2023). It also allows us to treat cases in which the DM may choose a vector of actions, the rewards from which are a quadratic function of the action vector and the external state in various periods; the problem of action choice to maximize expected reward in such a case is equivalent to a problem of minimizing a quadratic function of the DM's error in forecasting certain linear combinations of the value of the external state at various horizons.⁴

We suppose that DM produces estimates of $\tilde{z}_t$ without perfect knowledge of the first moment of the stochastic process for the external state. Instead, the parameter μ is assumed to be drawn from a prior distribution:

$$\mu \sim N(0, \Omega).$$

To simplify our discussion, we assume that the second moment of the stochastic process for the external state is known (more precisely, that the DM's decision rule can be optimized for particular values of these parameters, that are assumed to be the correct ones). Thus, the values of the parameters ρ and σ_ϵ^2 are assumed to be known, while μ is not. Conditional on the value of μ , the initial lagged state y_{-1} is assumed to be drawn from the prior distribution $N(\mu, \sigma_y^2)$, the ergodic distribution for the external state given a value for μ . When we consider the optimality of a possible decision rule for the DM, we integrate over this prior distribution of possible values for μ and y_{-1} , assuming that the decision rule must operate in the same way regardless of which values happen to be true in a given environment.

Importantly, we suppose that DM does not have perfect access to the past realizations of $(y_0, \dots, y_{t-1})$ at time t , though observing the current realization y_t without an error. The DM's decision rule must then respond adaptively to evidence about the unknown mean value based on a coarser information set, regardless of the assumed memory limitations. Conditioning on the DM's cognitive state at time t (and noting that this includes precise awareness of the value of y_t), the optimal estimate of $\tilde{z}_t$ will be given by

$$(1.3) \quad \sum_{j=0}^{\infty} A_j [\hat{\mu}_t + \rho^j (y_t - \hat{\mu}_t)],$$

where $\hat{\mu}_t$ is the DM's estimate of the unknown mean μ at each date t , based on the information available at the time that z_t must be chosen, and A_j is the vector with its i th element equal to $a_{i,j}$.

Note that in the absence of any memory limitation (and given the assumption of perfect observability of the realizations of y_t) it should be possible eventually for the DM to learn the value of μ to arbitrary precision, so that despite our assumption the value of μ must be learned, the optimal decision rule should coincide asymptotically

⁴For example, in a standard consumption-smoothing problem with quadratic consumption utility, the DM's level of expected utility depends on the accuracy with which "permanent income" is estimated at each point in time. This requires the DM to forecast a single variable $\tilde{z}_n$ for which the coefficient A_j is proportional to β^j for all $j \geq 0$.

with the full-information rational expectations prediction. We show, however, that this is not true when the precision of memory is bounded.

We show in the online Appendix that the DM's expected loss is not reduced by restricting attention to a class of decision rules of the form (1.3), under different possible assumptions about how the estimate $\hat{\mu}_t$ is formed.⁵ In the case of any forecasting rule of that kind, the loss function (1.2) is equal to

$$(1.4) \quad \alpha \cdot \sum_{t=0}^{\infty} \beta^t MSE_t,$$

plus a term that is independent of the DM's forecasts, where

$$MSE_t \equiv E[(\hat{\mu}_t - \mu)^2]$$

is the mean squared error in estimating μ , and $\alpha > 0$ is a constant that depends on the coefficients $\{A_j\}$ and W . Thus, one can equivalently formulate the DM's problem as one of optimal choice of an estimate $\hat{\mu}_t$ each period, so as to minimize the discounted sum of the current and future MSE_t .

B. A Model of Imprecise Memory

Feasible Memory Structures.—We assume that the memory carried into each period $t \geq 0$ can be summarized by a vector. We denote this memory state as a vector m_t of dimension d_t . We further assume that it is only through this memory state that DM can access information about past states. Thus, given the realized m_t , the action chosen in period t (i.e., the choice of $\hat{\mu}_t$) must be a function of the cognitive state specified by $s_t = (m_t, y_t)$. As noted above, we assume that current external state y_t is perfectly observable.⁶

We suppose that the future memory state evolves from the current cognitive state according to a linear law of motion of the form

$$(1.5) \quad m_{t+1} = \Lambda_t s_t + \omega_{t+1}, \quad \omega_{t+1} \sim N(0, \Sigma_{\omega, t+1}),$$

starting from an initial condition of dimension $d_0 = 0$ (i.e., s_0 consists only of y_0). Note that information about past states can be stored for future use through memory states. In other words, information in s_t not stored in m_{t+1} will not be accessible in periods following $t + 1$. Furthermore, the noise ω_{t+1} prevents the future memory state m_{t+1} from accurately reproducing the cognitive state s_t . Given this core structure, the proposed memory evolution is quite flexible. The dimension of the memory state is assumed only to be finite and is not required to be the same for all t . And the elements of the matrices Λ_t and $\Sigma_{\omega, t+1}$ are allowed to be arbitrary; we require only that $\Sigma_{\omega, t+1}$ must be positive semi-definite (though it need not be of full rank).

The linear-Gaussian memory evolution proposed in (1.5) nests a large class of memory structures proposed in the literature. The case of perfect memory can be accommodated by our notation, by assuming that Λ_t is a $(t+1) \times (t+1)$ identity

⁵ See online Appendix A for details of the argument.

⁶ The case in which the current state is observable only imprecisely is discussed in Sung (2024).

matrix and $\Sigma_{\omega,t+1} = O$ so that $d_t = t$ and the elements of the vector m_t correspond to the values $(y_0, y_1, \dots, y_{t-1})$. Another type of memory that we can consider is one in which only the n most recent past observations of the external state can be recalled, though these are recalled with perfect precision. The requirement that forecasts be functions of the cognitive state would then require them to be functions of $(y_{t-n}, \dots, y_{t-1}, y_t)$ for some finite n , as under the hypothesis of “natural expectations” proposed by Fuster, Hébert, and Laibson (2011). This case would correspond to a specification in which $d_t = n$ for all t ; Λ_t is an $n \times (n + 1)$ matrix, the right $n \times n$ block of which is an identity matrix, and the first column of which consists entirely of zeros; and $\Sigma_{\omega,t+1} = O$. In Section VIB, we discuss other proposals for memory structure that our formalism allows to consider, such as an “episodic” memory of the kind assumed by Neligh (2024). However, none of those specifications turns out to be optimal.

Imprecision of Memory.—We limit the precision of memory by further assuming that there is a cost of storing and/or accessing the memory state. Since the memory state m_{t+1} provides information about the cognitive state s_t , we suppose that the cost is an increasing and convex function of the Shannon mutual information between the two random variables.⁷ Thus, denoting $\mathcal{I}_t$ to be the mutual information between m_{t+1} and s_t , the memory cost in period t , $c(\mathcal{I}_t)$, is increasing and convex in $\mathcal{I}_t$.⁸

Such limitation in the memory system can be viewed either as a cost of *storing* information in memory or *retrieving* information from memory. In the first interpretation, only the information content $\mathcal{I}_t$ (about the period t cognitive state) is recorded in memory, which is available with perfect precision in the following period. In the second interpretation, the period t cognitive state is perfectly recorded in memory, but only the information content $\mathcal{I}_t$ is retrieved in the following period. In our proposed memory evolution (1.5), these two formulations are identical. This is because of our assumption that only the signal m_{t+1} that is retrieved in period $t + 1$ can be stored for future use; thus, only the fidelity with which the retrieved memory m_{t+1} reproduces the cognitive state s_t matters.⁹

We can now formulate the dynamic optimization problem with which we are concerned. We suppose that the optimal memory structure (1.5), together with the rule for choosing an estimate $\hat{u}_t$ as a function of each period’s cognitive state, minimizes the total discounted costs

$$(1.6) \quad \sum_{t=0}^{\infty} \beta^t [\alpha \cdot MSE_t + c(\mathcal{I}_t)].$$

⁷Mutual information is a nonnegative scalar quantity that can be defined for any joint distribution for (s_t, m_{t+1}) , that measures the degree to which the realized value of either random variable provides information about the value of the other (Cover and Thomas 2005). This measure is used to determine the relative cost of different information structures in the rational inattention theory of Sims (2003); properties of this measure as an information cost function are discussed in Caplin, Dean, and Leahy (2022). Our use of it to measure the cost of more precise memory follows Fox and Tishby (2012).

⁸We assume that the cost of memory precision is additively separable across periods; this facilitates our use of dynamic programming methods. Fox and Tishby (2012) instead assume a finite upper bound on $\sum_t \mathcal{I}_t$ (in a finite-horizon problem). This is related (though not identical) to the case $c(\mathcal{I}_t) = \mathcal{I}_t$ in our formalism (treated below), which also implies perfect substitutability of precision costs between periods.

⁹Under the retrieval-cost interpretation, our model remains importantly different from the one proposed by Afrouzi et al. (2023). The recursive nature of the memory states in our model is crucial in predicting the “recency bias.” See Section VIB for further discussion.

This objective takes into account the cost of less accurate forecasts (1.4) and the cost of greater memory precision. Note that no expectation is needed in (1.6), since both $MS E_t$ and $\mathcal{I}_t$ are functions of the entire joint probability distribution of possible values for μ , m_t , y_t , $\hat{\mu}_t$ and m_{t+1} .

Implications of the Linear-Gaussian Memory Structure.—For any memory state m_t that can be retrieved from the law of motion (1.5), the posterior distribution over possible values of $(\mu, y_{-1}, y_0, \dots, y_{t-1})$ implied by m_t will be a multivariate Gaussian distribution. Thus, a finite set of first and second moments fully characterize the posterior associated with the memory state. Importantly, the particular memory state m_t at a given date t can be identified by the associated vector of first moments. This is because the second moments of the posterior are the same for any possible memory state at any time t : they depend on the matrices $\{\Lambda_\tau, \Sigma_{\omega, \tau+1}\}$ for $\tau < t$, but not on the history of the external state, or on the history of realizations of the memory noise $\{\omega_{t+1}\}$. It follows that we can label memory states using the vector of first moments associated with that memory state; we use this notation in writing explicit dynamics for the memory state in the following sections.

Among the state variables about which the memory state may convey information, we are particularly interested in what the memory state implies about the value of μ . As mentioned above, the posterior distribution for the parameter μ implied by any realized m_t will be a Gaussian distribution. Furthermore, following an observation of y_t , the posterior distribution for μ implied by the cognitive state s_t again will be a Gaussian distribution. We introduce the notation

$$(1.7) \quad \mu|s_t \sim N(\hat{\mu}_t, \hat{\sigma}_t^2).$$

Note that $\hat{\sigma}_t^2$ is necessarily positive (complete certainty about the value of μ cannot be achieved in finite time, even in the case of perfect memory), and must satisfy the upper bound

$$(1.8) \quad \hat{\sigma}_t^2 \leq \hat{\sigma}_0^2 \equiv \frac{\Omega \sigma_y^2}{\Omega + \sigma_y^2},$$

where $\hat{\sigma}_0^2$ corresponds to the degree of uncertainty about μ after observing the external state once in the case of no informative memory whatsoever (the DM's situation in period $t = 0$).

From (1.7), we observe that the DM's optimal decision rule will be $\hat{\mu}_t$ each period, and that the minimum achievable value for the loss function (1.4), given the memory structure, will be $\sum_{t=0}^{\infty} \beta^t [\alpha \cdot \hat{\sigma}_t^2]$. It follows that the optimal memory structure will be the one that minimizes the implied value of

$$(1.9) \quad \sum_{t=0}^{\infty} \beta^t [\alpha \cdot \hat{\sigma}_t^2 + c(\mathcal{I}_t)].$$

The Role of Sequential Updating in Our Model.—We study the imprecision of memory in a situation where observations arrive sequentially and new forecasts are produced after each observation. When making forecasts, we assume that the DM accesses past observations only through memory. Our model's prediction for the DM's forecasts after observing the data $\{y_0, y_1, \dots, y_t\}$ would be quite different if

one supposed that these data were all presented simultaneously at the time when the DM must produce a forecast. In the latter case, in the absence of any imprecision in the observation of current data, the DM's forecast should correspond to the ideal Bayesian forecast (RE forecast). If instead we were to assume a cost of precise observation of the data, but a mutual information cost of the kind hypothesized by Sims (2003), then the model would predict insensitivity of the DM's forecast to the data (relative to the RE forecast), but equal insensitivity to each of the observations: there would be no "recency effect" of the kind predicted by our model, as discussed in Section IID below.

This aspect of our model's predictions is in line with experimental literature showing that inferences from a series of observations are indeed different depending whether the data are presented simultaneously or sequentially (e.g., Shanteau, 1970). Recency effects are particularly prevalent in studies in which data are presented sequentially with an updated estimate elicited after each additional piece of information ("step-by-step" responses), as opposed to studies in which an estimate is elicited only after the complete sequence of evidence has been presented ("end-of-sequence" responses); see Hogarth and Einhorn (1992).¹⁰

II. The Optimal Memory Structure: A Simple Example

We first suppose that the external state y_t is an independent draw each period from a Gaussian distribution, $y_t \sim N(\mu, \sigma_y^2)$; that is, $\rho = 0$ in (1.1). We provide analytical results about the optimal dimension and structure of the memory system and the kind of conclusions that follow from it. In the succeeding sections, we show that results from this section remain qualitatively similar in a more general setup.

A. Integrating Memory with New Observations

For any memory state m_t realized in period t , we first introduce the notation

$$\mu|m_t \sim N(\bar{m}_t, \Sigma_t)$$

for the conditional distribution for μ implied by m_t . After the value of y_t is observed, these beliefs are updated to a posterior of the form (1.7), whose first moment is characterized as

$$(2.1) \quad \hat{\mu}_t = \bar{m}_t + \gamma_t(y_t - \bar{m}_t),$$

using the notation

$$(2.2) \quad \gamma_t = \frac{\Sigma_t}{\Sigma_t + \sigma_y^2}$$

¹⁰We do not here present a model of the type of decision problem in which no response is required until the end of a sequence of observations. The differing attentional biases reported in this case may indicate that the cost of accessing the contents of a cognitive state at some date t at a later date $t + j$ depends on the number of occasions during the interim in which the information has had to be retrieved and integrated with other (newer) information before being stored again, rather than simply reflecting the passage of time. We leave this further complication for analysis elsewhere.

for the “Kalman gain,” and the second moment is

$$(2.3) \quad \hat{\sigma}_t^2 = \frac{\Sigma_t \sigma_y^2}{\Sigma_t + \sigma_y^2}.$$

In the case of perfect memory, the DM’s forecast z_t can be a function of the complete sequence of observations $(y_0, \dots, y_t)$ from some initial period zero onwards. Then, the computation of the MMSE estimate of μ is a standard Kalman filtering problem. Since we can identify $(\bar{m}_t, \Sigma_t)$ with $(\hat{\mu}_{t-1}, \hat{\sigma}_{t-1}^2)$, the equations (2.1) and (2.3) can be solved recursively to determine $\hat{\mu}_t$ and $\hat{\sigma}_t^2$ for all t (given a sequence of realizations of the state), starting from initial conditions $\hat{\mu}_{-1} = 0$, $\hat{\sigma}_{-1}^2 = \Omega$. These equations imply that the precision $\hat{\sigma}_t^{-2}$ grows linearly with the number of observations, and hence that $\hat{\sigma}_t^2 \rightarrow 0$ as $t \rightarrow \infty$, regardless of the sequence of observations. Eventually, the correct value of μ is learned to arbitrary precision, and new observations cease to affect the estimate of μ ($\gamma_t \rightarrow 0$), and consequently cease to affect the DM’s forecast. Thus, forecasts are eventually the same as under an assumption of (full information) rational expectations. We wish to examine how these conclusions change in the case of imperfect memory.

B. Optimal Memory Structure

In the simple case considered in this section, the only aspect of the cognitive state s_t that is worth remembering later is what was known at t about the value of μ . This is because when $\{y_t\}$ is an i.i.d. random variable, the only possible relevance of memory for decisions in periods $t + 1$ or later is the evidence that memory can provide about the value of the parameter μ .¹¹ The parameters of the distribution (1.7) capture this information. Furthermore, we can see that the first moment of this distribution, $\hat{\mu}_t$, is the only aspect of the cognitive state that needs to be remembered. This is because the second moment $\hat{\sigma}_t^2$ is independent of the history of realizations of the external state and hence the same in all possible cognitive states s_t , following the linear-Gaussian dynamics in our model.¹² Importantly, any other aspect of s_t that is not correlated with $\hat{\mu}_t$ is information that is not needed but is costly to remember. In other words, any such aspect of s_t is of no relevance to the estimates of μ , but a memory state m_{t+1} that is informative about it requires a higher value of $\mathcal{I}_t$ without increasing the information provided about the value of μ . Thus, an optimal memory structure will make the distribution of the random variable $m_{t+1}|s_t$ a function only of $\hat{\mu}_t$.

Moreover, the information about past knowledge that the memory state needs to convey can be completely summarized by the implied value of $\bar{m}_t$. It follows that we can label the memory state carried into any period t using the implied value of $\bar{m}_t$, and from here on we use the notation $\bar{m}_t$ to refer to the memory state.¹³ It follows

¹¹ We generalize the argument to the case of a persistent process in Section III.

¹² It depends on t , which is to say the number of observations that have occurred; but this is assumed to be available as an input to the decision rule, rather than something that has to be remembered using costly memory.

¹³ Here we refer only to our analysis of the simple case treated in this section. The notation used below for the case with $\rho \neq 0$ is slightly different.

that the optimal memory state is one-dimensional, and the dynamic evolution of this state can furthermore be given a representation of the form

$$(2.4) \quad \bar{m}_{t+1} = \lambda_t \hat{\mu}_t + \bar{\omega}_{t+1},$$

where λ_t is a scalar between $[0, 1]$ and $\bar{\omega}_{t+1}$ is a scalar random variable that is independently drawn from $N(0, \sigma_{\bar{\omega}, t+1}^2)$, a distribution with the variance $\sigma_{\bar{\omega}, t+1}^2 = \lambda_t(1 - \lambda_t)(\Omega - \hat{\sigma}_t^2)$. This memory structure implies that the DM's prior about μ at the start of period $t + 1$ is

$$(2.5) \quad \mu|m_{t+1} \sim N(\bar{m}_{t+1}, \Sigma_{t+1}),$$

where $\bar{m}_{t+1}$ is given by (2.4) and the prior uncertainty Σ_{t+1} is given by

$$(2.6) \quad \Sigma_{t+1} = (1 - \lambda_t)\Omega + \lambda_t\hat{\sigma}_t^2.$$

Thus, the precision of the DM's memory and of the knowledge of μ at each point in time is fully captured by a sequence $\{\lambda_t\}$.

The representation above clarifies the role of imprecise memory in shifting the DM's memory-based prior beliefs about μ away from its perfect-memory counterpart. In particular, we can see how the beginning-of-period prior $\mu|m_{t+1}$ is different from $\mu|s_t$, which would have been the DM's prior under the perfect memory assumption. From (2.4), we can see that the prior mean $\bar{m}_{t+1}$ shrinks toward the unconditional prior mean (which is assumed to be zero) and contains "memory noise" $\bar{\omega}_{t+1}$ that is uncorrelated with s_t . From (2.6), we also see that the prior uncertainty Σ_{t+1} is bigger than the posterior uncertainty of the previous period $\hat{\sigma}_t^2$. Note also that in the limit as $\lambda_t \rightarrow 1$, we approach the case of a perfect memory, in which $\mu|m_{t+1}$ is identical $\mu|s_t$.

Finally, in order for the quantity $\bar{m}_{t+1}$ specified by (2.4) to correspond to the conditional expectation $E[\mu|m_{t+1}]$, it is necessary that

$$\sigma_{\bar{\omega}, t+1}^2 = \lambda_t(1 - \lambda_t)\text{var}[\hat{\mu}_t] = \lambda_t(1 - \lambda_t)[\Omega - \hat{\sigma}_t^2].$$

Thus, the complete dynamics (2.4) of the memory state are specified by giving a numerical value for the scalar λ_t . Since $\sigma_{\bar{\omega}, t+1}^2$ should be a nonnegative scalar for a memory structure to be feasible, λ_t must be a quantity between 0 and 1. It then remains only to show how this quantity is determined.

C. Determination of the Sequence $\{\lambda_t\}$

The memory cost at time t depends on $\mathcal{I}_t$, the mutual information between m_{t+1} and s_t . According to the optimal memory structure in (2.4), it is straightforward to see that the only information about s_t contained in m_{t+1} is the information about the value of $\hat{\mu}_t$. Hence, the value of $\mathcal{I}_t$ will be the mutual information between the random variables $\hat{\mu}_t$ and $\bar{m}_{t+1}$. It follows that

$$\mathcal{I}_t = -\frac{1}{2}\ln(1 - \lambda_t),$$

which is an increasing function of λ_t . The optimal value of λ_t is then determined by a trade-off between the memory cost $c(\mathcal{I}_t)$, which is increasing in λ_t , and the losses resulting from entering period $t + 1$ with a greater degree of uncertainty Σ_{t+1} , which is decreasing in λ_t as a result of (2.6).

In this section, we consider a specific functional form for $c(\mathcal{I}_t)$ such that for a given scalar $\bar{\mathcal{I}}$, memory cost takes the following form:

$$c(\mathcal{I}_t) = \begin{cases} 0, & \text{if } \mathcal{I}_t \leq \bar{\mathcal{I}} \\ \infty, & \text{otherwise} \end{cases}.$$

In other words, we assume that there is a finite upper bound $\bar{\mathcal{I}}$ on the rate of information transmission, so that feasible memory structures must satisfy the constraint $\mathcal{I}_t \leq \bar{\mathcal{I}}$.¹⁴ Because $\mathcal{I}_t$ is an increasing function of λ_t , this constraint can alternatively be expressed as a constraint $\lambda_t \leq \bar{\lambda}$, where $0 < \bar{\lambda} < 1$. (The limiting case in which $\bar{\lambda} = 1$ corresponds to no upper bound on the mutual information, and hence perfect memory.) We can further show that the uncertainty about the value of μ in all periods $\tau > t$ is minimized (and hence the loss function (1.9) is minimized) by setting λ_t as large as possible, consistent with the constraint. Hence in each period the upper bound constraint will bind, and the optimal memory structure will satisfy

$$(2.7) \quad \lambda_t = \bar{\lambda}, \forall t.$$

Equations (2.1)–(2.3), (2.4)–(2.6), and (2.7) then constitute a complete system of equations to recursively determine the evolution of the variables $\{\bar{m}_t, \Sigma_t, \hat{\mu}_t, \hat{\sigma}_t^2\}$ for all $t \geq 0$ given the sequence of observations $\{y_t\}$, starting from initial conditions (corresponding to the unconditional prior) $\bar{m}_0 = 0, \Sigma_0 = \Omega$. This generalizes the standard Kalman filter dynamics to allow for noisy memory.

D. Characteristics of the Forecast Dynamics

In the simple problem considered here, the optimal forecast each period is given by $z_t = \hat{\mu}_t$; the predictable part of the forecast error (if any) will simply be the predictable error (if any) in $\hat{\mu}_t$ as an estimate of μ ; and the mean squared error of the forecast will equal $\hat{\sigma}_t^2 + \sigma_y^2$, where $\hat{\sigma}_t^2$ is the mean squared error of the estimate of μ . Thus, we need only analyze the dynamics of the estimate $\hat{\mu}_t$ and the estimation error that this reflects.

An important implication of noisy memory is that the estimation error $\hat{\sigma}_t^2$ is not fully eliminated even after an arbitrarily long period of learning. This prediction holds even though the coefficients of the data generating process are assumed not to change. From equations (2.3) and (2.6), we can see that $\hat{\sigma}_t^2$ does not converge to zero as long as $\bar{\lambda} < 1$ and instead converges to a positive number and that this long-run limiting value depends on the unconditional prior variance Ω . This is in contrast to the perfect memory case, where the value of μ is correctly learned to arbitrary precision, regardless of the size of Ω .

¹⁴In Section III, we generalize this assumption to allow $\mathcal{I}_t$ to be increased at a finite marginal cost.

Because our model implies that a DM does not learn the true value of μ with certainty even in the long run, it follows that the DM's forecasts can be quite different from rational expectations forecasts—that is, the forecasts of an ideal statistician who knows the true coefficient values. From the standpoint of an observer who is able to determine the true process, the forecasts of the DM with limited memory will appear to be systematically biased. The biases in the DM's forecasts will furthermore fluctuate over time, in response both to variations in the external state (to which the DM reacts differently than someone with rational expectations would) and to noise in the evolution of the memory state.

Below we discuss two notable characteristics of the forecast dynamics. The conclusions that we draw remain qualitatively similar in the more general case considered in the next section, though the calculations required are more complex in that case.

Stationary Fluctuations in the Long Run.—We obtain a particularly simple characterization of the systematic pattern of forecast biases if we consider the long run. The predictions of the equations (2.2), (2.3), (2.6) and (2.7) for very large values of t imply that Σ_t , $\hat{\sigma}_t^2$ and γ_t all converge to constant values. In particular, we use a notation $\gamma_\infty < 1$ to denote the long-run limit of the Kalman gain (that remains positive for any $\bar{\lambda} < 1$).¹⁵ In this case, our model, like the model of “natural expectations” of Fuster, Laibson, and Mendel (2010) and Fuster, Hébert, and Laibson (2011), predicts a stationary pattern of forecast biases that are not a result of incomplete learning.

In the long run, equations (2.1), (2.4) and (2.7) imply that the memory state $\bar{m}_t$ evolves according to an AR(1) process with Gaussian innovations,

$$(2.8) \quad \bar{m}_{t+1} = \rho_m \bar{m}_t + \bar{\lambda} \gamma_\infty y_t + \bar{\omega}_{t+1},$$

where the constant coefficient $\rho_m \equiv \bar{\lambda}(1 - \gamma_\infty)$ measures the intrinsic persistence of the memory state. For any bound $\bar{\lambda} < 1$, we necessarily have $0 < \rho_m < 1$ and $\sigma_{\bar{\omega}}^2 > 0$, so that (2.8) describes a stationary autoregressive process for beliefs. The two independent exogenous sources of variation in these dynamics are the innovations in y_t and the memory noise $\bar{\omega}_{t+1}$.

The DM's optimal estimate $\hat{\mu}_t$ at each point in time will then be a linear function of $\bar{m}_t$ and y_t , with coefficients that are also time-invariant. This allows us to analyze not only the extent to which the DM's forecasts should differ from rational expectations forecasts, but the correlation that one should observe between the bias in the DM's forecasts and other observable variables. In particular, the biases in the DM's forecasts will be correlated with the evolution of the external state. An unexpectedly high observed value for y_t will be interpreted (because of the DM's uncertainty about μ) as implying a higher estimate of μ , and this increase in the DM's estimate of μ will furthermore persist, decaying only gradually in subsequent periods.

“Recency Bias” in Expectation Formation.—One type of systematic difference between observed expectations and those of a perfect Bayesian decision-maker that

¹⁵ We give a closed-form solution for the value of γ_∞ as a function of the exogenous model parameters in online Appendix F.

has often been reported is “recency bias” (e.g., Malmendier et al. 2016)—a tendency for expectations to be influenced by more-recent observations, even when in principle, observations of a given time series at earlier dates should be equally relevant as a basis for inference. Our model predicts that such a bias should exist, as a consequence of optimal adaptation to limited memory precision (or to the cost of maintaining a more precise memory). Observations of the external state farther in the past are recalled with more noise, and as a consequence are given less weight in estimating parameters of the data generating process than would be optimal in the case of a perfect memory of past data.

The law of motion of the memory state (2.8) implies that, in the case that data have been generated in accordance with this law of motion for a sufficiently long time, we can express the value of the memory state $\bar{m}_{t+1}$ as a function of the sequence of external states $\{y_\tau\}$ and the sequence of memory noise realizations $\{\bar{\omega}_{\tau+1}\}$ for $\tau \leq t$:

$$(2.9) \quad \bar{m}_{t+1} = \bar{\lambda} \gamma_\infty \sum_{j=0}^{\infty} (\rho_m)^j y_{t-j} + \bar{\omega}_{t+1}^{sum},$$

where $\bar{\omega}_{t+1}^{sum} \equiv \sum_{j=0}^{\infty} (\rho_m)^j \bar{\omega}_{t+1-j}$ is a serially correlated Gaussian noise term, independent of the history of observations $\{y_{t-j}\}$. The DM’s estimate of the mean μ can then be written in the form

$$(2.10) \quad \hat{\mu}_t = \sum_{j=0}^{\infty} \alpha_j y_{t-j} + (1 - \gamma_\infty) \bar{\omega}_t^{sum},$$

where $\alpha_j = \gamma_\infty (\rho_m)^j$. Because the limiting coefficients α_j are positive, the estimate $\hat{\mu}_t$ (and hence the DM’s forecast z_t) continues to be influenced by recent observations y_{t-j} even when t is large—unlike the rational expectations forecast, which would be $\hat{\mu}_t = \mu$. Thus, the DM’s forecast is predicted to “overreact” to news about recent observations.¹⁶ Furthermore, our model predicts decreasing weights on observations made farther in the past. This is a notable difference between our model and the one proposed by Afrouzi et al. (2023).

Note also, the forecasts specified by (2.10) are similar to those implied by the constant-gain variant model of least squares learning about an unknown mean (Evans and Honkapohja 2001). We provide, however, a justification for the declining weight on observations farther in the past, as a consequence of optimal forecasting based on an imperfect memory, and furthermore endogenize the value of the “gain parameter.” We discuss the similarities and differences between our model and the constant gain learning in more detail in Section VI.

III. Generalizing the Analysis

We now consider a more general class of linear-quadratic decision problems, allowing for persistent dynamics in the external state $\{y_t\}$ and for a more general specification of the cost of memory precision.

Note first, forecasting future values y_{t+h} requires more information about past observations than simply the information they convey about the value of μ , when y_t

¹⁶We compare the predictions of our model to the measures of overreaction reported by Afrouzi et al. (2023) in Section V.

is a persistent process: the DM further needs an estimate of the previous realization y_{t-1} . This change has consequences for the information content of the optimal memory system. It is useful to introduce the notation

$$x_t \equiv (\mu, y_{t-1})'$$

to refer to the relevant state vector for forecasting at time t . As before, the memory state m_{t+1} conveys information about what the cognitive state s_t revealed about the external state vector x_t ; but now x_t does not consist solely of the mean μ .

As discussed in Section I, any memory state m_t realized from the law of motion (1.5) implies that $x_t|m_t$ is described as a multivariate Gaussian distribution. We can generalize the notation used in the i.i.d. case and write

$$x_t|m_t \sim N(\bar{m}_t, \Sigma_t).$$

Note that now $\bar{m}_t$ is a vector with two elements, and is not just the posterior mean estimate of μ . The posterior mean and variance for μ after observing y_t will then be given by the usual Kalman filter formulas:

$$(3.1) \quad \hat{\mu}_t = e_1' \bar{m}_t + \gamma_{1t} [y_t - c' \bar{m}_t],$$

$$(3.2) \quad \hat{\sigma}_t^2 = e_1' \Sigma_t e_1 - \gamma_{1t}^2 [c' \Sigma_t c + \sigma_\epsilon^2],$$

where the vectors are defined as $e_1' \equiv [1 \ 0]$ and $c' \equiv [1 - \rho \ \rho]$.¹⁷ The Kalman gain is equal to¹⁸

$$(3.3) \quad \gamma_{1t} = \frac{e_1' \Sigma_t c}{c' \Sigma_t c + \sigma_\epsilon^2}.$$

As in the previous section, we label the memory state carried into any period t using the implied value of $\bar{m}_t$ and refer to it as the memory state. Below we discuss how $\bar{m}_t$ and Σ_t are determined under the optimal memory system.

A. Optimal Memory Structure

We derive the optimal evolution of the memory state, given the linear law of motion of the memory state (1.5) and the associated memory costs $c(\mathcal{I}_t)$ presented in Section I. We present our results in the main text and leave the detailed derivations in online Appendix C.

When the external state vector x_t consists more than the mean μ , we show that the optimal memory state evolves according to the following representation.

$$(3.4) \quad \bar{m}_{t+1} = \bar{\Lambda}_t \underbrace{E[x_{t+1}|s_t]}_{\equiv \bar{s}_t} + \bar{\omega}_{t+1},$$

¹⁷ Here the vector e_1' selects μ from x_t , and c' allows us to write $y_t = c'x_t + \epsilon_{yt}$.

¹⁸ We use a 1 subscript in the notation for this variable because it is the first element of a vector of Kalman gains, defined in the more general formula given in online Appendix B.

where $E[x_{t+1}|s_t] = (\hat{\mu}_t, y_t)'$, which we denote with a notation $\bar{s}_t$. As the memory state encodes information about a vector $\bar{s}_t$ (instead of just $\hat{\mu}_t$ as in the i.i.d. case), the loading $\bar{\Lambda}_t$ is now a 2×2 matrix. Likewise, the memory noise $\bar{\omega}_{t+1}$ is now a 2×1 random vector drawn from $N(O, \Sigma_{\bar{\omega}, t+1})$, a distribution with the variance $\Sigma_{\bar{\omega}, t+1} = (I - \bar{\Lambda}_t) \text{var}[\bar{s}_t] \bar{\Lambda}_t'$ where I is a 2×2 identity matrix. This memory structure will imply that conditional on the realized memory state, DM's belief about next-period state vector is

$$(3.5) \quad x_{t+1}|m_{t+1} \sim N(\bar{m}_{t+1}, \Sigma_{t+1}),$$

where $\bar{m}_{t+1}$ is given by (3.4) and Σ_{t+1} is given by

$$(3.6) \quad \Sigma_{t+1} = (I - \bar{\Lambda}_t) \Sigma_0 + \bar{\Lambda}_t \text{var}[x_{t+1}|s_t],$$

where Σ_0 is the prior uncertainty about x_t in the case of no informative memory.

Note that DM's posterior uncertainty about μ fully describes the 2×2 matrix $\text{var}[x_{t+1}|s_t]$: $\hat{\sigma}_t^2$ is on the (1,1) element and all the other elements are zero (as DM observes the realized y_t without an error). Thus, we can define $\mathcal{L}(\hat{\sigma}_t^2)$ to be the set of feasible $\bar{\Lambda}_t$, so that the resulting Σ_{t+1} is a proper variance-covariance matrix (i.e., symmetric and p.s.d). A sequence of $\{\bar{\Lambda}_t\}$ satisfying this feasibility condition fully describes how precise DM's memory of x_t is at all t . This conclusion is analogous to the i.i.d. case where a sequence of a scalar loading $\lambda_t \in [0, 1]$ determines the precision of the DM's memory of μ .

According to (3.4)–(3.6), the value of $\hat{\sigma}_t^2$ in any period t is implied by the choice of memory structure for the periods *prior* to t , and the value of $\hat{\sigma}_{t+1}^2$ is uniquely determined by $\bar{\Lambda}_t \in \mathcal{L}(\hat{\sigma}_t^2)$. We denote this function as

$$\hat{\sigma}_{t+1}^2 = f(\hat{\sigma}_t^2, \bar{\Lambda}_t)$$

for any nonnegative $\hat{\sigma}_t^2$ satisfying the bound (1.8) and any $\bar{\Lambda}_t \in \mathcal{L}(\hat{\sigma}_t^2)$. Then given that we start from an initial degree of uncertainty $\hat{\sigma}_0^2$ at time $t = 0$ defined by (1.8), we can define the class of sequences $\{\bar{\Lambda}_t\}$ for all $t \geq 0$ with the property that $\bar{\Lambda}_t \in \mathcal{L}(\hat{\sigma}_t^2)$ for all $t \geq 0$; let us call this class $\mathcal{L}^{seq}$. Moreover, for any sequence of transition matrices in $\mathcal{L}^{seq}$, we can uniquely define the sequence of values $\{\hat{\sigma}_t^2\}$ for all $t \geq 0$ implied by it. Thus, given any sequence $\{\bar{\Lambda}_t\} \in \mathcal{L}^{seq}$, we can calculate the implied sequence of losses $\{MSE_t\}$ from forecast inaccuracy.

B. Determination of $\{\bar{\Lambda}_t\}$

As we discuss in online Appendix D, the evolution of the memory state (3.4)–(3.6) implies that the mutual information between m_{t+1} and s_t is given by

$$(3.7) \quad \mathcal{I}_t = \mathcal{I}(\bar{\Lambda}_t) = -\frac{1}{2} \log \det(I - \bar{\Lambda}_t),$$

and the feasibility condition $\bar{\Lambda}_t \in \mathcal{L}(\hat{\sigma}_t^2)$ ensures that $\det(I - \bar{\Lambda}_t) \in (0, 1]$. Thus, the mutual information $\mathcal{I}_t$ is well-defined and necessarily nonnegative. As the elements of $\bar{\Lambda}_t$ are made small, so that memory ceases to be very informative about

the prior cognitive state, $I - \bar{\Lambda}_t$ approaches the identity matrix, and $\mathcal{I}_t$ approaches zero. If $\bar{\Lambda}_t$ is varied in such a way as to make one of its eigenvalues approach 1, $I - \bar{\Lambda}_t$ approaches a singular matrix, and $\Sigma_{\hat{\omega}, t+1}$ must approach a singular matrix as well; this means that in the limit, some linear combination of the elements of $\bar{s}_t$ is a random variable with positive variance that comes to be recalled with perfect precision. In this case, $\det(I - \bar{\Lambda}_t)$ approaches zero, so that $\mathcal{I}_t$ grows without bound.

Thus, a given sequence of transition matrices $\{\bar{\Lambda}_t\}$ uniquely determines sequences $\{MSE_t, \mathcal{I}_t\}$, allowing the value of the objective (1.6) to be calculated. The problem of optimal design of a memory structure can then be reduced to the choice of a sequence $\{\bar{\Lambda}_t\} \in \mathcal{L}^{seq}$ so as to minimize (1.6). This objective is necessarily well-defined for any such sequence, since each of the terms is nonnegative; the infinite sum will either converge to a finite value, or will diverge, in which case the sequence in question cannot be optimal.¹⁹

A Recursive Formulation.—We now observe that if for any point in time t , we know the value of $\hat{\sigma}_t^2$ (which depends on the choices made regarding memory structure in periods $\tau < t$), the set of admissible transition matrices $\{\bar{\Lambda}_\tau\}$ for $\tau \geq t$ specifying the memory structure from that time onward will depend only on the value of $\hat{\sigma}_t^2$, and not any other aspect of choices made about the earlier periods. Moreover, any admissible continuation sequence $\{\bar{\Lambda}_\tau\}$ for $\tau \geq t$ implies unique continuation sequences $\{MSE_\tau, \mathcal{I}_\tau\}$ for $\tau \geq t$, so that the value of the continuation objective

$$(3.8) \quad \sum_{\tau=t}^{\infty} \beta^{\tau-t} [\alpha \cdot MSE_\tau + c(\mathcal{I}_\tau)]$$

will be well-defined.²⁰

We can then consider the problem of choosing an admissible continuation plan $\{\bar{\Lambda}_\tau\}$ for $\tau \geq t$ so as to minimize (3.8), given an initial condition for $\hat{\sigma}_t^2$. (This is simply a more general form of our original problem choosing memory structures for all $t \geq 0$ to minimize (1.6), given the initial condition for $\hat{\sigma}_0^2$ specified in (1.8).) Let $V(\hat{\sigma}_t^2)$ be the lowest achievable value for (3.8), as a function of the initial condition $\hat{\sigma}_t^2$. This function is defined for any value of $\hat{\sigma}_t^2$ satisfying the bound (1.8), and is independent of the date t from which we consider the continuation problem. Note that the lower bound necessarily lies in the interval

$$(3.9) \quad \alpha \hat{\sigma}_t^2 \leq V(\hat{\sigma}_t^2) \leq \alpha \left[\hat{\sigma}_t^2 + \frac{\beta}{1-\beta} \hat{\sigma}_0^2 \right].$$

(The lower bound follows from the fact that $MSE_t = \hat{\sigma}_t^2$, and all other terms in (3.8) must be nonnegative; the upper bound is the value of (3.8) if one chooses $\bar{\Lambda}_\tau = 0$ for all $\tau \geq t$, which is among the admissible continuation plans.)

¹⁹Note, it is clearly possible to choose memory structures for which the infinite sum converges. For example, if one chooses $\bar{\Lambda}_t = 0$ for all $t \geq 0$ (perfectly uninformative memory at all times), $MSE_t = \hat{\sigma}_0^2$ and $\mathcal{I}_t = 0$ each period, and (1.6) will equal the finite quantity $(\alpha \hat{\sigma}_0^2)/(1-\beta)$.

²⁰Since a finite value for the continuation objective is always possible (see (3.9)), it is clear that plans making (3.8) a divergent series cannot be optimal and can be excluded from consideration.

This value function also necessarily satisfies a Bellman equation of the form

$$(3.10) \quad V(\hat{\sigma}_t^2) = \min_{\bar{\Lambda}_t \in \mathcal{L}(\hat{\sigma}_t^2)} \left[\alpha \hat{\sigma}_t^2 + c(\mathcal{I}(\bar{\Lambda}_t)) + \beta V(f(\hat{\sigma}_t^2, \bar{\Lambda}_t)) \right],$$

where $\mathcal{I}(\bar{\Lambda}_t)$ is the function defined in (3.7). Thus, once we know how to compute the value function for arbitrary values of $\hat{\sigma}_{t+1}^2$, the problem of the optimal choice of a memory structure in any period t can be reduced to the one-period optimization problem stated on the right-hand side of (3.10). This indicates how the memory structure for period t must be chosen to trade off the cost $c(\mathcal{I}_t)$ of retaining a more precise memory against the continuation loss $V(\hat{\sigma}_{t+1}^2)$ from having access to a less precise memory in period $t + 1$.

Let $\mathcal{F}$ be the class of continuous functions $V(\hat{\sigma}_t^2)$, defined for values of $\hat{\sigma}_t^2$ consistent with (1.8), and consistent with the bounds (3.9) everywhere on this domain. Then (3.10) defines a mapping $\Phi : \mathcal{F} \rightarrow \mathcal{F}$: given any conjectured function $V(\hat{\sigma}_{t+1}^2) \in \mathcal{F}$ that is used to evaluate the right-hand side for any value of $\hat{\sigma}_t^2$, the minimized value of the problem on the right-hand side defines a new function $\tilde{V}(\hat{\sigma}_t^2)$ that must also belong to $\mathcal{F}$. Condition (3.10) states that the value function that defines the minimum achievable continuation loss must be a fixed point of this mapping: a function such that $V = \Phi(V)$.

We can further show that for any function $V \in \mathcal{F}$, the function $\Phi(V)$ defined by the right-hand side of (3.10) is necessarily a monotonically increasing function.²¹ It follows that the fixed point $V(\hat{\sigma}_t^2)$ must be a monotonically increasing function. Moreover, we can restrict the domain of the mapping Φ to the subset $\mathcal{F}^*$ of increasing functions.

This then provides us with an approach to computing the optimal memory structure for a given parameterization of our model. First, we find the value function $V(\hat{\sigma}_t^2) \in \mathcal{F}^*$ that is a fixed point of the mapping Φ , by iterating Φ to convergence. Then, given the value function, we can numerically solve the minimization problem on the right-hand side of (3.10) to determine the optimal transition matrix $\bar{\Lambda}_t$ in any period, once we know the value of $\hat{\sigma}_t^2$ for that period. Solution of this problem also allows us to determine the value of $\hat{\sigma}_{t+1}^2 = f(\hat{\sigma}_t^2, \bar{\Lambda}_t)$, so that the entire sequence of values $\{\hat{\sigma}_\tau^2\}$ for all $\tau \geq t$ can be determined once we know $\hat{\sigma}_t^2$. Finally, we recall that for the initial period $t = 0$, the value of $\hat{\sigma}_0^2$ is given by (1.8); we can thus solve for the entire sequence $\{\hat{\sigma}_t^2\}$ for all $t \geq 0$ by integrating forward from this initial condition.

Once we have determined the sequence of values $\{\hat{\sigma}_t^2\}$ implied by an optimal memory structure for each period, we can determine the sequence of $\{\gamma_{1,t}, \Sigma_t\}$, using (3.3) and (3.6). This in turn allows us to calculate the DM's optimal estimate $\hat{\mu}_t$ each period, as a function of the history of realizations $\{y_\tau\}$ of the external state for all $0 \leq \tau \leq t$ and the history of realizations of the DM's memory noise $\{\bar{\omega}_{\tau+1}\}$ for all $0 \leq \tau \leq t - 1$, using (3.1) and (3.4). The DM's complete vector of forecasts z_t each period is then given by (1.3).

²¹ See online Appendix E.1 for a proof.

The Optimality of a One-Dimensional Memory State.—We can show further that the optimal memory state must have a one-dimensional representation. This simplifies the computational formulation of the optimization problem on the right-hand side of (3.10), and provides further insight into the nature of an optimally imprecise memory. Although the information contained in the cognitive state s_t that is relevant for predicting (at time t) what actions will be desirable for the DM in later periods is two-dimensional (both elements of $\bar{s}_t$ matter, if $\rho > 0$, and except when memory is completely uninformative, these are not perfectly correlated with each other), we find that it is optimal for the DM's memory to include only a noisy record of a single linear combination of the two variables. Moreover, this is true regardless of how small memory costs may be. The details of the argument are in online Appendix E.2.

For $\bar{m}_t$ to have one-dimensional representation, it must be that the loading matrix $\bar{\Lambda}_t$ is not full rank. Specifically, we show that the optimal $\bar{\Lambda}_t$ at each t takes the following particular form

$$(3.11) \quad \bar{\Lambda}_t = \lambda_t \text{var}[\bar{s}_t] v_t v_t',$$

where λ_t is a scalar between $[0, 1]$ and v_t is a vector normalized to satisfy $v_t' \text{var}[\bar{s}_t] v_t = 1$. Accordingly, the optimal memory state in (3.4) is represented as

$$(3.12) \quad \bar{m}_{t+1} = \text{var}[\bar{s}_t] v_t \tilde{m}_{t+1},$$

where the scalar quantity $\tilde{m}_{t+1}$ is defined as

$$(3.13) \quad \tilde{m}_{t+1} = \lambda_t v_t' \bar{s}_t + \tilde{\omega}_{t+1}, \quad \tilde{\omega}_{t+1} \sim N(0, \lambda_t(1 - \lambda_t)).$$

Thus, the optimal memory state $\bar{m}_{t+1}$ is spanned from a uni-variate memory variable $\tilde{m}_{t+1}$. The choice of memory structure can then be reduced to a problem of choosing a scalar quantity λ_t and the direction of a vector v_t in each period t .

There is in fact a fairly simple intuition for this result. Note that in any period t , the optimal estimate of the unknown value of μ will be linear in the elements of m_t , as shown in (3.1). In other words, the only information in the memory state m_t that matters for the estimate $\hat{\mu}_t$ is the particular linear combination of m_t (that is determined by the Kalman filter). This single quantity sufficiently captures the information memory reveals for DM's estimation task.

In addition, we show that the choice of memory structure can be further reduced to a problem of only choosing λ_t . We show that there is a uniquely optimal v_t for any given values $0 \leq \hat{\sigma}_t^2 \leq \sigma_0^2$ and $0 < \lambda_t \leq 1$ ²². This is because of the monotonicity of the value function $V(\hat{\sigma}_{t+1}^2)$; the optimal weight vector v_t in any period must be the one that minimizes the resulting $\hat{\sigma}_{t+1}^2$ for a given pair of $\hat{\sigma}_t^2$ and λ_t . Given the explicit solution for the optimal v_t in online Appendix E.3, we can write the law of motion

$$\hat{\sigma}_{t+1}^2 = \bar{f}(\hat{\sigma}_t^2, \lambda_t)$$

²²Note that no solution is needed in the case that $\lambda_t = 0$, since in this case v_t is undefined.

starting from the initial condition $\hat{\sigma}_0^2$ defined in (1.8). (The function is well-defined when $\lambda_t = 0$ since $\bar{f}(\hat{\sigma}_t^2, 0) = \hat{\sigma}_0^2$.) Thus, we can solve for the dynamics of $\{\hat{\sigma}_t^2\}$ implied by any sequence $\{\lambda_t\}$. Moreover, it follows from (3.7) that the mutual information associated with the period t memory structure will be given by

$$(3.14) \quad \mathcal{I}_t = -\frac{1}{2} \log(1 - \lambda_t),$$

just as in the i.i.d. case discussed in Section II. The Bellman equation (3.10) can therefore be written in the simpler form

$$(3.15) \quad V(\hat{\sigma}_t^2) = \min_{0 \leq \lambda_t \leq 1} \left[\alpha \hat{\sigma}_t^2 + c \left(-\frac{1}{2} \log(1 - \lambda_t) \right) + \beta V(\bar{f}(\hat{\sigma}_t^2, \lambda_t)) \right].$$

Thus, the dynamic optimization problem reduces to finding an optimal value for a scalar variable $\lambda_t \in [0, 1]$ for a given state variable $\hat{\sigma}_t^2$. A more precise memory, captured by a higher λ_t , yields lower continuation loss at $t + 1$ at the expense of higher cost at t . We turn now to the features of the model solution to the dynamic optimization problem just posed.

IV. Features of the Model Solution

Here we provide numerical examples of solutions for an optimal memory structure, under alternative assumptions about both the degree of persistence of the process that must be forecasted and the nature of the memory cost function. In particular, we consider two functional forms for the memory cost. The first kind is introduced in Section II, in which we suppose that the capacity of information flow between two consecutive time periods is given. In this case, the accuracy of the memory system is exogenously fixed. In the other case, we suppose that the amount of information stored in memory can be increased at a cost. Thus, the memory cost linearly increases with the accuracy of the memory system. We discuss how the model solutions vary with the severity of the memory cost, using these two functional forms for the memory cost.

In reporting our results, it is useful to describe the model solution in terms of scale-invariant quantities—that is, ones that are independent of the value of σ_y , indicating the amplitude of the transitory fluctuations in the external state around its mean. Thus, we parameterize the degree of prior uncertainty about μ not in terms of a value for Ω (the variance of the prior distribution for μ), but rather by a value for $K \equiv \Omega/\sigma_y^2$ (the variance of the prior distribution for μ/σ_y). We similarly measure the degree of uncertainty about μ conditional on the cognitive state at a given point in time (i.e., after a given amount of experience) not in terms of the value of $\hat{\sigma}_t^2$, but rather by the scaled uncertainty measure $\eta_t \equiv \hat{\sigma}_t^2/\sigma_y^2$. In terms of this scaled uncertainty measure, an optimal memory structure minimizes the value of

$$\sum_{t=0}^{\infty} \beta^t [\eta_t + \tilde{c}(\mathcal{I}_t)],$$

a scaled version of (1.6), where the scaled cost function is defined as $\tilde{c}(\mathcal{I}_t) \equiv c(\mathcal{I}_t)/(\alpha \sigma_y^2)$.²³

Our scale-invariant model is then completely specified by values for the parameters ρ, β, K and the scaled cost function $\tilde{c}(\mathcal{I}_t)$. In our quantitative analysis, we assume that each “period” of our discrete-time model corresponds to a quarter of a year (the variable to be forecasted is a quarterly time series), and hence set $\beta = 0.99$ (implying a discount rate of 4 percent per annum). We consider a variety of values $0 \leq \rho < 1$ for the assumed degree of serial correlation of the external state, and explore the effects of different assumptions regarding the degree of prior uncertainty and the size of information costs.

A. The Case of a Fixed per Period Bound on Mutual Information

We begin by considering the case in which $\tilde{c}(\mathcal{I}_t) = 0$ for all $\mathcal{I}_t \leq \bar{\mathcal{I}}$, but values of $\mathcal{I}_t$ greater than $\bar{\mathcal{I}}$ are infeasible, as assumed in Section II. Solution for the optimal memory structure is particularly simple in this case. Because of (3.14), the per period bound on mutual information can equivalently be written as an upper bound $\lambda_t \leq \bar{\lambda}$, just as in Section II. The optimal memory structure in period t is then characterized by the λ_t that minimizes $\bar{f}(\hat{\sigma}_t^2, \lambda_t)$ subject to this constraint. We show in the online Appendix²⁴ that the minimizing value of λ_t is necessarily the largest feasible value; hence in the solution to this problem, $\lambda_t = \bar{\lambda}$, the value determined by the per period information bound.

The dynamics of the uncertainty measure are then given by $\hat{\sigma}_{t+1}^2 = \bar{f}(\hat{\sigma}_t^2, \bar{\lambda})$. In terms of the rescaled variables, the law of motion can be written as

$$(4.1) \quad \eta_{t+1} = \phi(\eta_t; \bar{\lambda}),$$

where $\phi(\eta; \bar{\lambda})$ is a function that is independent of the scale factor σ_y .²⁵ For any value of $\bar{\lambda}$ indicating the tightness of the constraint on the complexity of memory, equation (4.1) indicates how the DM’s degree of uncertainty about μ evolves as additional observations of the external state are made. Starting from the initial condition $\eta_0 = K/(K+1)$ implied by (1.8), the law of motion (4.1) can be iterated to obtain a unique solution for the complete sequence of values $\{\eta_t\}$ for all $t \geq 0$. In the limiting case $\bar{\lambda} = 1$ (unlimited memory), the law of motion (4.1) takes the especially simple form

$$(4.2) \quad \frac{1}{\eta_{t+1}} = \frac{1}{\eta_t} + \frac{1-\rho}{1+\rho}.$$

This is simply the standard result for the linear growth in posterior precision under Bayesian updating as additional observations are made; it has the implication that η_t declines monotonically, and converges to zero for large t . Thus, in the case of perfect

²³ Dividing by α further reduces the numbers of parameters that we need to specify in considering the different possible forms that the optimal memory structure may take, since it is only the relative weights on the two loss terms in the objective (1.6) that matter for the optimal memory structure.

²⁴ See online Appendix G.1 for details of the argument.

²⁵ See online Appendix G.1 for an explicit algebraic solution for this function.

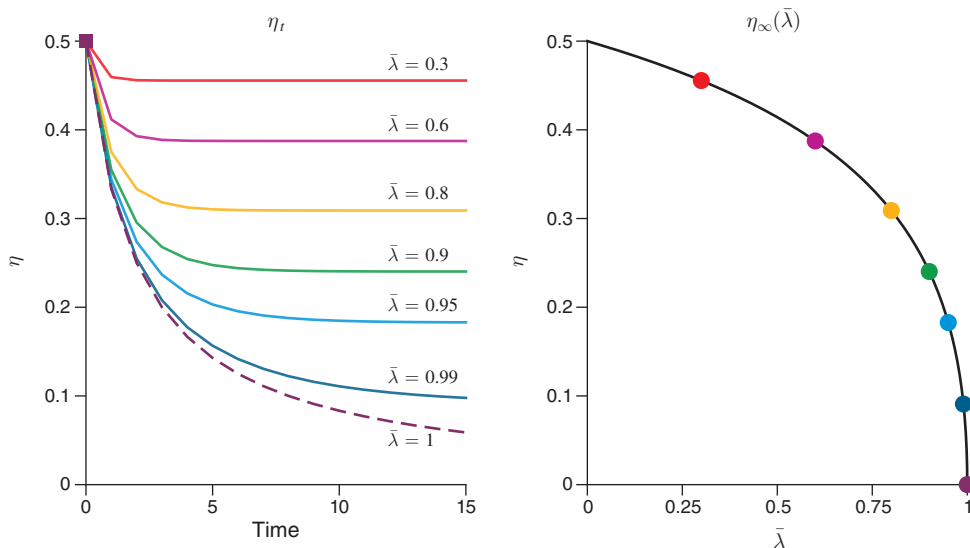


FIGURE 1. THE EVOLUTION OF UNCERTAINTY ABOUT μ (FOR VARYING DEGREES OF $\bar{\lambda}$)

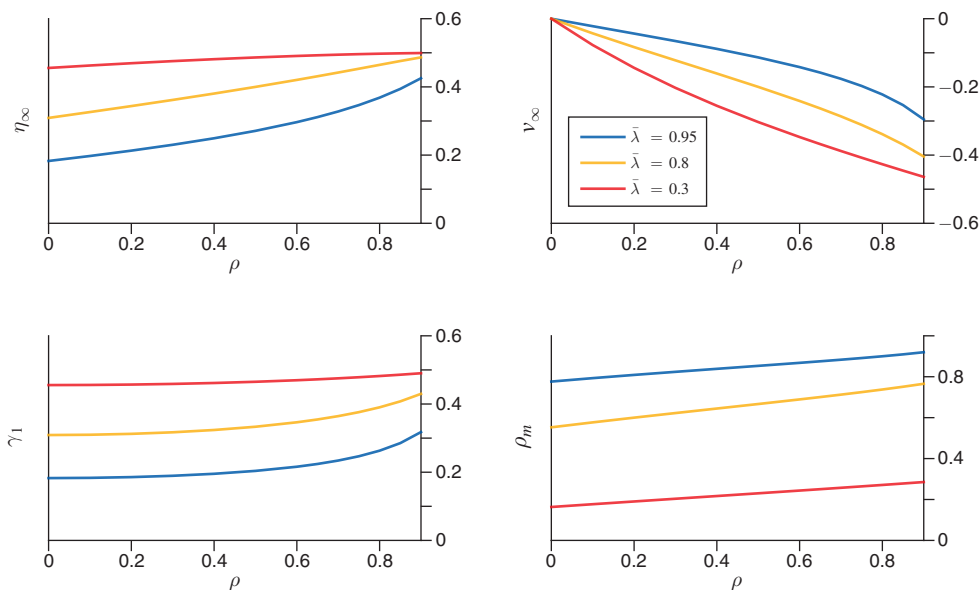
Notes: The evolution of scaled uncertainty about μ as the number t of previous (imperfectly remembered) observations grows. The right panel shows the long-run value of scaled uncertainty (to which η_t converges as $t \rightarrow \infty$) as a function of the constraint on the complexity of memory, parameterized by $\bar{\lambda}$.

memory, the DM should eventually learn the value of μ with perfect precision, and hence make forecasts of the kind implied by the hypothesis of rational expectations.

When $\bar{\lambda} < 1$, instead, the law of motion (4.1) implies that η_t should decrease initially, as even imprecise memory of the DM's observations of the external state reduces uncertainty to some degree, but that η_t remains bounded away from zero, and converges to a value $\eta_\infty(\bar{\lambda}) > 0$. This is illustrated in Figure 1, which shows the dynamics implied by (4.1) for each of several different values of $\bar{\lambda}$, in the case that $\rho = 0$ and $K = 1$.²⁶ The left panel plots the sequence of values $\{\eta_t\}$ implied by (4.1) for a given value of $\bar{\lambda}$. (Note that the initial value η_0 is the same in each case.) The right panel shows the value η_∞ to which the sequence converges as t grows; this value depends on $\bar{\lambda}$, and the functional relationship between $\bar{\lambda}$ and this limiting degree of uncertainty can be described by a function $\eta_\infty(\bar{\lambda})$, plotted as a smooth curve in the right panel of the figure.

In the case that $\bar{\lambda} = 1$ (shown as a dashed curve in the left panel of Figure 1), the sequence $\{\eta_t\}$ decreases monotonically to zero at the rate predicted by the difference equation (4.2). But for any number of prior observations $t > 0$, the value of η_t remains higher the lower is $\bar{\lambda}$ (that is, the tighter the memory constraint), and the long-run degree of uncertainty about μ , measured by η_∞ , is a decreasing function of $\bar{\lambda}$ as well, as shown by the curve in the right panel of the figure. Because of the limit on the amount of information that can be retained in memory, the DM's uncertainty

²⁶The effects of variation in the parameters ρ and K are illustrated in additional figures shown in online Appendix H.1.

FIGURE 2. THE OPTIMAL MEMORY STRUCTURE IN THE LONG RUN (FOR VARYING DEGREES OF $\bar{\lambda}$)

Notes: Coefficients describing the optimal memory structure in the long run, as a function of the degree of persistence ρ of the external state, for alternative values of $\bar{\lambda}$. Respective panels show the long-run values for η_t (measuring uncertainty about μ), the direction vector v_t (indicating the content of the memory state), the Kalman gain γ_{1t} (for updating the DM's estimate of μ), and ρ_m (measuring the intrinsic persistence of fluctuations in the memory state).

about the value of μ never falls below a certain level, even in the long run, despite our assumption that the value of μ is fixed for all time. We further observe that the long-run degree of uncertainty η_∞ is larger, the smaller is $\bar{\lambda}$ (that is, the tighter the constraint on memory). In the limit as $\bar{\lambda}$ approaches zero (corresponding to a constraint that memory must be completely uninformative), the long-run degree of uncertainty η_∞ approaches the prior degree of uncertainty $\eta_0 = K/(K + 1)$.

As η_t falls along one of these trajectories, the weight vector v_t that constitutes the optimal memory state (3.13) shifts as well. As η_t converges to the long-run value η_∞ , the optimal weight vector v_t similarly converges to a long-run value v_∞ , indicating the particular linear combination of $\hat{\mu}_t$ and y_t that is imprecisely recorded in memory each period. Associated with this stationary long-run memory structure there will also be a stationary long-run value for the Kalman gain coefficient γ_{1t} in equation (3.1), and more generally, stationary values for the coefficients of the linear difference equations describing the joint dynamics $\{y_t, \hat{m}_t\}$ of the external state and the memory state.

These long-run stationary coefficients will depend on the value of $\bar{\lambda}$ and also on the value of ρ . Figure 2 indicates how variation in each of these parameters affects several of the long-run stationary coefficients.²⁷ In each panel, a curve shows how the coefficient in question varies as a function of ρ (for values of ρ between 0.0 and

²⁷ See online Appendix H.1 for the formulas used to calculate each of the coefficients plotted here as functions of the model parameters.

0.9), for a given value of $\bar{\lambda}$; curves of this kind are shown for each of three different values of $\bar{\lambda}$, ranging between $\bar{\lambda} = 0.95$ (in which case memory is relatively precise) and $\bar{\lambda} = 0.30$ (in which case it is much more constrained). All of the curves shown in Figure 2 are again for the case of prior uncertainty $K = 1$.

The upper-right panel of the figure shows the long-run direction vector v_∞ ; the quantity reported on the vertical axis is the long-run value of the ratio v_{2t}/v_{1t} of the vector's two components.²⁸ Thus, a value of -0.3 for this quantity means that the univariate memory state $\tilde{m}_{t+1}$ is (up to a multiplicative factor that does not affect its information content) equal to the value of $\hat{\mu}_t - 0.3 y_t$, plus additive Gaussian noise. The figure shows that when $\rho = 0$, the optimal univariate memory state involves $v_2 = 0$; that is, only the current estimate $\hat{\mu}_t$ of the unknown mean is remembered with noise, with the current observation y_t being completely forgotten. This is optimal because when $\rho = 0$, the current value y_t contains no information that is relevant for improving subsequent forecasts of the external state, except to the extent that it helps to improve the DM's estimate of μ (which information is already reflected in the estimate $\hat{\mu}_t$). Instead, when the external state is serially correlated, it is optimal to commit to memory a linear combination of $\hat{\mu}_t$ and the current state y_t ; in the case that $\rho > 0$, the optimal linear combination puts a negative relative weight on y_t , to an extent that is greater the greater the degree of serial correlation, and greater the tighter the constraint on memory.

The upper-left panel of the figure shows the long-run degree of uncertainty about μ , measured by η_∞ . As shown in Figure 1, when $\rho = 0$, η_∞ is a decreasing function of $\bar{\lambda}$. We see in Figure 2 that this is also true when $\rho > 0$. However, for a given memory constraint $\bar{\lambda}$, the long-run value η_∞ is also an increasing function of ρ , with the degree of increase when the external state is highly persistent being particularly notable when memory is more accurate. The greater the serial correlation of the state, the fewer the effective number of independent noisy observations of μ that the DM receives over any finite time period. Thus, even under perfect Bayesian updating, equation (4.2) indicates that the rate at which precision is increased by each additional observation is smaller the larger is ρ . In the case of perfect memory, the long-run degree of uncertainty about μ is nonetheless zero (there is simply slower convergence to that long-run value when ρ is large); but with moderately imperfect memory, the effective amount of experience that can ever be drawn upon remains bounded, so that the uncertainty about μ remains larger forever when ρ is larger. When memory is even more imperfect, not much more than one observation (the most recent one) can be used in any event, so that the value of η_∞ is in this case less sensitive to the value of ρ .

In the long run, the dynamics of the cognitive state $\bar{s}_t$ and the memory state $\bar{m}_{t+1}$ are described by linear equations with constant coefficients. The lower-left panel of Figure 2 shows the long-run value for the Kalman gain γ_{1t} in (3.1). With imperfect memory, this is always a quantity between 0 and 1, meaning that a higher value of the current state y_t raises the DM's estimate of the value of μ , though by less than

²⁸This information (together with the value of η_∞ given in the upper left panel) suffices to completely determine the vector v_t , since the vector is normalized so that $v_t' \text{var}[\bar{s}_t] v_t = 1$. The value of λ_t (given by the constraint $\bar{\lambda}$), the matrix $\text{var}[\bar{s}_t]$ (determined by the value of η_∞), and the vector v_t then completely determine the long-run stationary elements of the matrix $\bar{\Lambda}_t$ (using (3.11)) and hence also of the matrix $\Sigma_{\bar{\omega}}$ (using $\Sigma_{\bar{\omega},t+1} = (I - \bar{\Lambda}_t) \text{var}[\bar{s}_t] \bar{\Lambda}_t$); thus the dynamics of the memory state given by (3.4) are completely specified.

the amount of the increase in y_t . For a given value of ρ , the Kalman gain is larger the tighter the constraint on memory; in the limit as $\bar{\lambda} \rightarrow 1$ (perfect memory), the long-run value of this coefficient approaches zero (as the true value of μ is eventually learned), while in the limit as $\bar{\lambda} \rightarrow 0$ (no memory), the value approaches a maximum value that is still less than one (because of the DM's finite-variance prior).

Finally, the lower-right panel reports the long-run value of ρ_m , a measure of the intrinsic persistence of the memory state. The impulse response function for the effect of a memory-noise innovation $\tilde{\omega}_t$ on the subsequent path of the univariate memory state $\tilde{m}_\tau$ is proportional to $(\rho_m)^{\tau-t}$ for all $\tau \geq t$;²⁹ thus, the value of ρ_m indicates the rate of exponential decay of the memory state back to its long-run average value. A smaller value of ρ_m means that the contents of memory decay more rapidly; for any value of ρ , the figure shows that ρ_m is smaller, the tighter the memory constraint. At the same time, while a larger value of ρ_m implies that memory persists for a longer time, it also implies that when memory noise creates an erroneous impression of prior experience, this bias in what is recalled about is also corrected more slowly. Thus, the value of ρ_m is an important determinant of the predicted persistence of forecast bias.

B. The Case of a Linear Cost of Information

Analysis of the model is more complex when instead the amount of information stored in memory each period can be increased at some finite cost. As an illustration we consider the polar opposite case in which $\tilde{c}(\mathcal{I}_t)$ is a linear function of $\mathcal{I}_t$, so that the marginal cost of a further increase in the mutual information is independent of how large it already is. Thus we assume that $\tilde{c}(\mathcal{I}_t) = \tilde{\theta} \cdot \mathcal{I}_t$, for some coefficient $\tilde{\theta} > 0$ which parameterizes the cost of memory.

In this case, the optimal choice of λ_t in any period will depend on the value of reducing uncertainty in the following period. We note that the value function $V(\hat{\sigma}_{t+1}^2)$ appearing in the Bellman equation (3.15) can be written as $\sigma_y \cdot \tilde{V}(\eta_{t+1})$, where η_{t+1} is the scaled uncertainty measure and the function $\tilde{V}(\eta)$ is independent of the scale factor σ_y (for given values of the parameters K, ρ, β and $\tilde{\theta}$). We can then write the Bellman equation in the scale-invariant form

$$(4.3) \quad \tilde{V}(\eta_t) = \min_{0 \leq \lambda_t \leq 1} \left\{ \eta_t - \frac{\tilde{\theta}}{2} \log(1 - \lambda_t) + \beta \tilde{V}(\phi(\eta_t; \lambda_t)) \right\}.$$

The optimal choice of λ_t in any period will be the value that solves the problem on the right-hand side of (4.3). This problem has a solution $\lambda_t = \lambda^*(\eta_t)$ which depends only on the value of η_t , the degree of uncertainty in period t determined by the memory structures chosen for periods prior to t .

Thus, we can solve for the optimal policy function $\lambda^*(\eta_t)$ once we know the value function $\tilde{V}(\eta_{t+1})$, and we can solve numerically for the value function by iterating

²⁹ Here we refer to the difference that the realization of $\tilde{\omega}_t$ makes for the forecasts of $\tilde{m}_\tau$ at different horizons $\tau \geq t$, by an observer who knows the true value of μ and the DM's cognitive state at time $t - 1$, in addition to observing the realization of $\tilde{\omega}_t$. See online Appendix H.1 for details of the calculation.

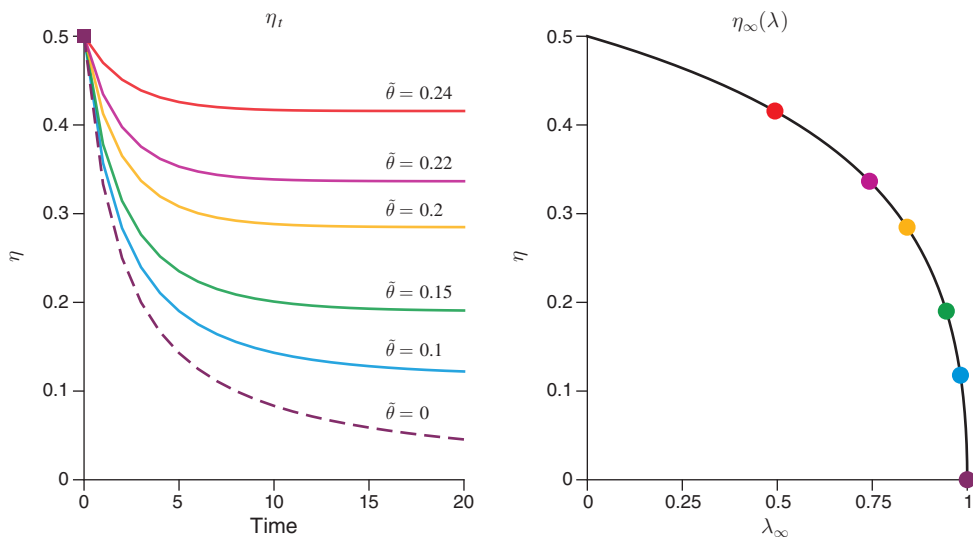


FIGURE 3. THE EVOLUTION OF UNCERTAINTY ABOUT μ (FOR VARYING DEGREES OF $\tilde{\theta}$)

Notes: The evolution of scaled uncertainty about μ as the number t of previous (imperfectly remembered) observations grows, now for the case of a linear cost of memory complexity. The right panel shows the long-run value of scaled uncertainty for each value of the cost parameter $\tilde{\theta}$, plotted as a point on the same locus of optimal long-run memory structures as in Figure 1.

the Bellman equation (4.3), as discussed further in the online Appendix.³⁰ The policy function $\lambda_t = \lambda^*(\eta_t)$ together with the law of motion

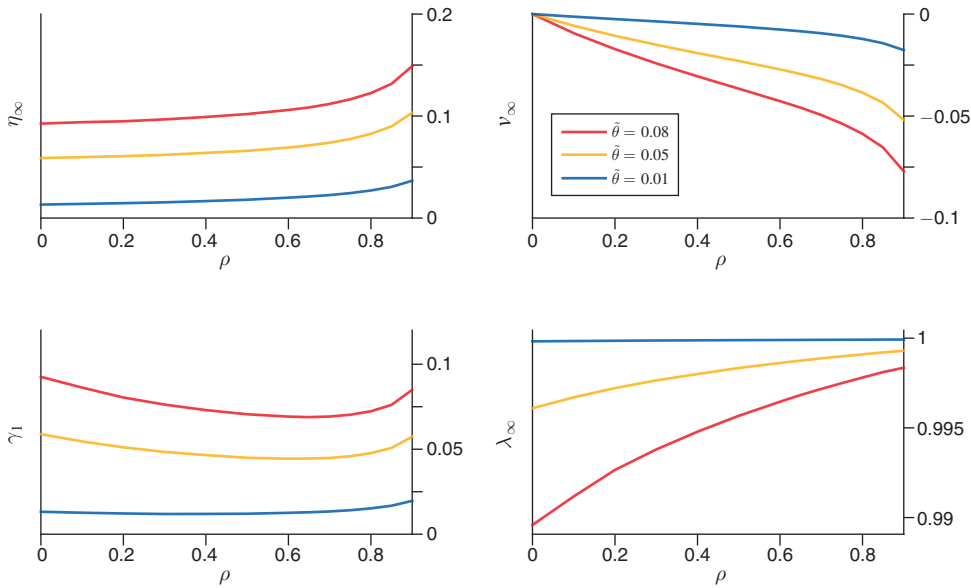
$$(4.4) \quad \eta_{t+1} = \phi(\eta_t; \lambda_t)$$

derived earlier can then be solved for the dynamics of the scaled uncertainty $\{\eta_t\}$ for all $t \geq 0$, starting from the initial condition $\eta_0 = K/(K+1)$.³¹ The dynamics of scaled uncertainty as a function of the number of observations t are shown for progressively larger values of $\tilde{\theta}$ in Figure 3, using the same format as in Figure 1. Once again, we see that while uncertainty about μ eventually falls to zero as a result of when there is no cost of memory complexity, as long as the cost is positive, the value of η_t remains bounded away from zero, and converges asymptotically to a value η_∞ that is higher the higher the cost of memory complexity.

Associated with such an asymptotic degree of uncertainty is a particular long-run memory structure $(\lambda_\infty, v_\infty)$, which will imply a particular long-run value for the Kalman gain γ_1 . The way in which the long-run values of these different quantities vary with different assumptions about the values of ρ and $\tilde{\theta}$ is illustrated in Figure 4. (We use the same convention as in Figure 2 to indicate the direction of the vector

³⁰ See online Appendix G.2 for details.

³¹ See online Appendix G.2 for further discussion of the implied dynamics.

FIGURE 4. THE OPTIMAL MEMORY STRUCTURE IN THE LONG RUN (FOR VARYING DEGREES OF $\tilde{\theta}$)

Notes: Coefficients describing the optimal memory structure in the long run, as a function of the degree of persistence ρ of the external state, in the case of a linear memory cost function, for alternative values of $\tilde{\theta}$. Respective panels show the long-run values for η , the direction vector ν , the Kalman gain γ_1 , and the memory precision coefficient λ .

ν_∞ in the upper-right panel of the figure.) As we vary ρ for a given value of $\tilde{\theta}$, the associated value of λ_∞ changes; hence the fixed- $\tilde{\theta}$ curves shown in Figure 4 do not correspond exactly to any of the fixed- λ curves plotted in Figure 2, even though each of the long-run memory structures associated with a pair $(\rho, \tilde{\theta})$ is identical to the long-run memory structure associated with some pair $(\rho, \bar{\lambda})$. As shown in the lower-right panel of the figure, the optimal λ_∞ rises as ρ increases, for any value of the cost parameter $\tilde{\theta} > 0$; the more persistent the external state that must be forecasted, the more it becomes worthwhile to pay a larger information cost in order to retain a more precise memory of prior experience.

Not surprisingly, we observe that for any value of ρ , increasing the memory cost $\tilde{\theta}$ makes it optimal for the long-run precision of memory λ_∞ to be smaller, and consequently for the long-run degree of uncertainty about μ to be larger. In the case of a sufficiently high value of $\tilde{\theta}$, it will be optimal for memory to be completely uninformative. In fact, this happens for a finite value of $\tilde{\theta}$, and it can occur abruptly, rather than through a gradual increase in the long-run degree of uncertainty η_∞ toward the limiting value of $\eta_0 = K/(K+1)$ as $\tilde{\theta}$ is increased. A graph of the relationship between η_∞ and the value of $\tilde{\theta}$ is shown in Figure 5, for the case $\rho = 0$, and two different possible values of K : $K = 1$ and $K = 10$. For each value of $\tilde{\theta}$, the value of η_∞ associated with the optimal memory structure is shown by a large blue dot.

In each panel of this figure, the continuous black curve is the correspondence consisting of all points $(\tilde{\theta}, \eta_\infty)$ such that η_∞ is a stationary solution of the Euler

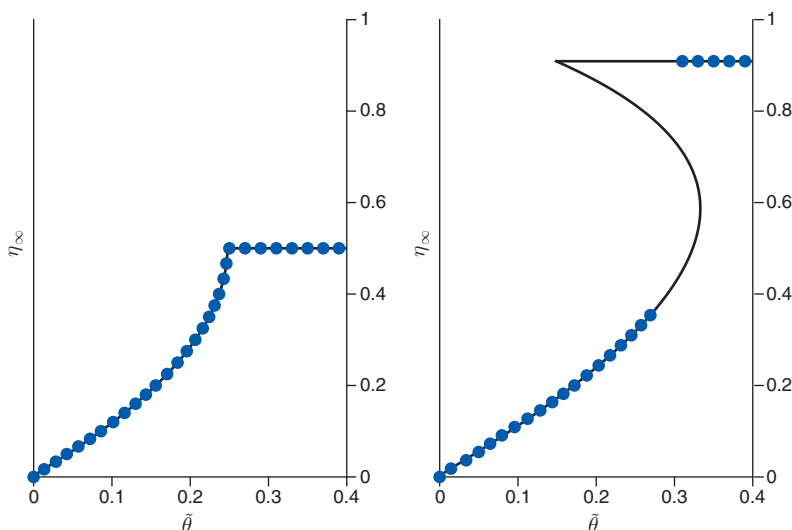


FIGURE 5. THE LONG-RUN UNCERTAINTY

Notes: Long-run value of the scaled uncertainty measure η_∞ (blue dots) as a function of the cost parameter $\tilde{\theta}$, in the case of a linear memory cost function. Left panel: $K = 1, \rho = 0$. Right panel: $K = 10, \rho = 0$.

equation associated with the optimization problem on the right-hand side of (4.3).³² The Euler equation represents a first-order condition for the optimal choice of the degree of precision of memory; satisfaction of this condition is necessary but not sufficient for memory precision leading to $\eta_{t+1} = \eta$ to be optimal starting from a situation in which $\eta_t = \eta$. Because the objective function on the right-hand side of (4.3) is not a convex function, it can have multiple local minima (as well as a local maximum located between two local minima). Which of the local minima represents the global minimum (and hence the optimal memory structure) can jump abruptly as a result of a small change in parameters.³³ This is what happens when the value of η_∞ changes abruptly in the right panel of Figure 5, for a value of $\tilde{\theta}$ slightly above 0.28.

In the $K = 10$ case, we see that there need not be a unique value of η_∞ for a given value of $\tilde{\theta}$ that represents a stationary solution to the Euler equation. For any value of $\tilde{\theta}$ greater than a critical value around 0.15, if one starts from $\eta_t = \eta_0$ (a completely uninformative memory), the choice of $\eta_{t+1} = \eta_0$ again represents a local minimum of the objective; hence $\eta = \eta_0$ is a stationary solution of the Euler equation for all of these values of $\tilde{\theta}$, as shown in the figure. However, for values of $\tilde{\theta}$ only moderately larger than the critical value (such as $\tilde{\theta} = 0.20$), this is not the only local minimum, and the global minimum is instead at an interior choice for λ_t . This value results in a path $\{\eta_t\}$ that converges to a different stationary value for η_∞ , on the lower branch of the correspondence (as shown for example by the blue dot for $\tilde{\theta} = 0.20$). Yet for values of $\tilde{\theta}$ that exceed a second critical value just above 0.28, the global minimum

³² See online Appendix G.4 for derivation of this equation.

³³ See online Appendix G.3 for a numerical example.

shifts from the interior minimum to the local minimum at $\eta_{t+1} = \eta_0$. For all values beyond this point, the optimal memory structure involves $\lambda_t = 0$ for all t , so that $\eta_\infty = \eta_0$ (as shown by the blue dots on the upper branch of the correspondence).

Thus, while the locus of fixed points $\eta_\infty(\lambda)$ is the same in Figures 1 and 3, all points on this locus represent possible long-run memory structures (attainable through an appropriate choice of $\bar{\lambda}$) in the case of a fixed upper bound on mutual information, but not all of them are always attainable in the case of a linear memory cost function. In the case $K = 1$, the two sets of long-run solutions are identical; but in the case $K = 10$, there is a range of values for η_∞ that are associated with particular (relatively low) values of $\bar{\lambda}$ but do not correspond to any possible value of $\hat{\theta}$.³⁴

C. Forecast Dynamics and Recency Bias

We can now generalize the discussion in Section II of the implied dynamics of the DM's forecasts in response to shocks to the external state. We first show that DM's cognitive state fluctuates around a steady state in the long run, just as in the i.i.d. case (2.8).

As shown in (3.13), the DM's memory regarding the state vector x_t can be summarized by a one-dimensional memory state $\tilde{m}_t$. The evolution of the vector $\tilde{s}_t \equiv (\tilde{m}_t, y_t)$ then fully characterizes the DM's forecasts of the future evolution of the external state. In the case that data have been generated in accordance with the law of motion of the memory state for a sufficiently long time for the coefficients of the DM's Kalman filter to converge, the dynamics of $\tilde{s}_t$ can be described by a VAR(1) system with constant coefficients of the form

$$(4.5) \quad \tilde{s}_{t+1} = f\mu + F\tilde{s}_t + u_{t+1}, \quad u_{t+1} \sim N(0, \Sigma_u),$$

where $u_{t+1} \equiv (\tilde{\omega}_{t+1}, \epsilon_{y,t+1})$, and f, F and Σ_u denote a two-vector and two 2×2 matrices of constant coefficients respectively. The matrix F furthermore has an upper-triangular form, while Σ_u is diagonal. We show in the online Appendix that the eigenvalues of the matrix F are ρ and ρ_m .³⁵ We further show that $0 < \rho_m < 1$, so that both y_t and $\tilde{m}_t$ exhibit stationary fluctuations around well-defined long-run average values which depend linearly on μ . The two independent exogenous sources of variation in this system are the innovations $\epsilon_{y,t+1}$ in the external state and the memory noise innovations $\tilde{\omega}_{t+1}$.

We can further see that DM's forecasts continue to exhibit recency bias, just as in the i.i.d. case (2.9). The system (4.5) implies that we can express the value of the memory state $\tilde{m}_{t+1}$ as a function of the sequence of external states $\{y_\tau\}$ for $\tau \leq t$ and the sequence of memory noise realizations $\{\tilde{\omega}_{\tau+1}\}$ for $\tau \leq t$:

$$(4.6) \quad \tilde{m}_{t+1} = F_{12} \cdot \sum_{j=0}^{\infty} (\rho_m)^j y_{t-j} + \tilde{\omega}_{t+1}^{sum},$$

³⁴We can show analytically that the continuous relationship shown in the left panel of Figure 5 occurs for all $K \leq 1$ when $\rho = 0$, while the backward-bending correspondence and consequent discontinuous relationship between $\hat{\theta}$ and η_∞ occurs for all $K > 1$. See online Appendix G.4 for further explanation.

³⁵See online Appendix H.1 for the derivation.

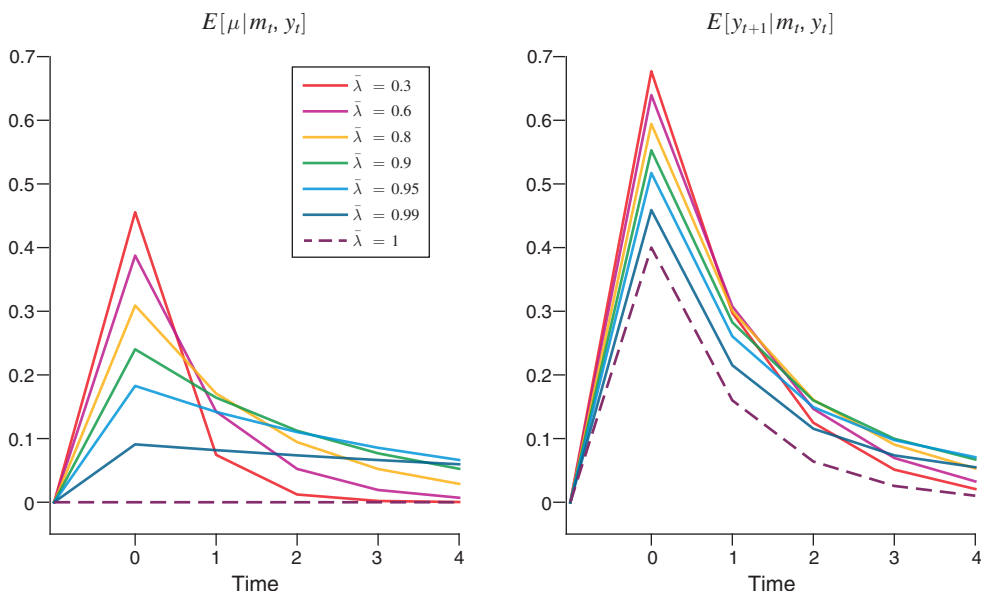


FIGURE 6. IMPULSE RESPONSE FUNCTIONS

Notes: Impulse responses of the DM's estimate of μ (left panel) and one-period-ahead forecast of the state (right panel) to a unit positive innovation in the observed value of y_t at the time marked as "time = 0" on the horizontal axis. Responses are plotted for alternative values of the information bound $\bar{\lambda}$, in the case that $K = 1$, $\rho = 0.4$.

where $\tilde{\omega}_{t+1}^{sum} \equiv \sum_{j=0}^{\infty} (\rho_m)^j \tilde{\omega}_{t+1-j}$ and F_{12} is the $(1, 2)$ element of the matrix F in (4.5).³⁶ Thus, the reduced memory state $\tilde{m}_t$ is more influenced by recent observations.

We can also use the law of motion (4.5) to trace out the impulse response of the vector $\tilde{s}_t$ (and hence of the DM's forecasts) to an innovation in the external state y_t . Figure 6 plots the responses when the precision of memory is bounded each period, in the case that $K = 1$ and $\rho = 0.4$.³⁷ The left panel shows how $\hat{\mu}_\tau$ responds in the long run to a unit positive innovation in the value of y_t . In the case that $\bar{\lambda} = 1$ (perfect memory), the value of μ is learned with precision. However, when memory is imperfect, an observation of a high value of y_t leads to a higher estimate $\hat{\mu}_t$ of the mean. This additional effect on the forecast (leading to overreaction relative to the RE benchmark) is initially larger the smaller is $\bar{\lambda}$; but a smaller $\bar{\lambda}$ (tighter memory constraint) also causes the additional effect to die out more rapidly, since its propagation can only be through the DM's memory of her previous judgment about the value of μ . The right panel shows the impulse response of the one-quarter-ahead forecast $\hat{y}_{\tau+1|\tau}$ to a unit positive innovation in y_t , using the same conventions as in the left panel. When $\rho > 0$, the rational expectations forecast (corresponding to $\bar{\lambda} = 1$ in the figure) is itself increased by a positive innovation in y_t (by an amount equal to fraction ρ of the innovation), and the increase in the forecast is furthermore persistent (decaying back

³⁶This is a stationary random process with a finite unconditional variance, since $0 < \rho_m < 1$, as shown in online Appendix H.1. Note that the fact that $\rho_m < 1$ implies that forecasts necessarily exhibit recency bias.

³⁷These impulse responses are plotted for a variety of assumptions about the degree of persistence of the external state in online Appendix H.1.

to its original level at a rate proportional to ρ^{T-t}). But when $\bar{\lambda} < 1$, the forecast is increased by even more, owing to the fact that the higher observation of y_t increases the DM's estimate of μ as well.

V. Experimental Evidence

We have shown that our model provides an explanation for important qualitative features of observed subjective expectations. Here we briefly discuss the model's quantitative fit with data on subjective expectations from the laboratory experiment of Afrouzi et al. (2023). We focus on this particular evidence for a quantitative test of our model, because it involves forecasts of a stationary AR(1) process, and in that sense matches exactly the problem assumed in our theoretical analysis above. A laboratory experiment also has the advantage over field studies of allowing us to be sure exactly what the true data generating process is, and exactly what information is available to decision-makers at each point in time (though of course questions remain about how the situation is understood by the experimental subjects, and what they pay attention to).

As noted in the introduction, Afrouzi et al. (2023) conduct a laboratory experiment in which subjects observe successive realizations of an AR(1) process, and forecast what the next realizations should be. They find that subjects' reported expectations overreact to innovations in this process, as predicted by our model (as well as the related model of noisy memory that they discuss). They give particular emphasis to a measure of overreaction in which a subject's forecast $\hat{y}_{t+h|t}$ (where h is the number of realizations in advance for which the forecast is solicited in trial t) is regressed on the realization of the variable just before the forecast is solicited:

$$(5.1) \quad \hat{y}_{t+h|t} = a_{s,h} + b_{s,h}y_t + v_t.$$

A separate regression (with coefficients $a_{s,h}, b_{s,h}$) can be estimated for each of several horizons h . Afrouzi et al. are interested in the difference between the "implied persistence" measured by the estimated value $\rho_h^s \equiv b_{s,h}^{1/h}$ and the corresponding coefficient ρ of the y_t process. The authors measure the degree of overreaction of expectations to news by the extent to which ρ_h^s is larger than ρ . Note that this is an example of a test of the predictability of forecast errors, since the coefficient of a regression of the forecast error $y_{t+h} - \hat{y}_{t+h|t}$ on y_t will equal $\rho^h - b_{s,h}$.

We can investigate what our model of expectation formation on the basis of an imperfect memory implies about the relationship between ρ_h^s and ρ in the case of a stationary AR(1) process. Here we consider the predicted values of the regression coefficients in the long run, as the length of the time series used to estimate them goes to infinity. Then, subjective forecasts should be given by

$$\hat{y}_{t+h|t} = (1 - \rho^h)\hat{\mu}_t + \rho^h y_t,$$

so that the predicted coefficient $b_{s,h}$ in regression (5.1) will equal

$$(5.2) \quad b_{s,h} = (1 - \rho^h)\beta_{\hat{\mu}|y} + \rho^h,$$

where $\beta_{\hat{\mu}|y}$ is the coefficient in a regression of $\hat{\mu}_t$ on y_t ,

$$\beta_{\hat{\mu}|y} = \frac{\text{cov}[\hat{\mu}_t, y_t | \mu]}{\text{var}[y_t | \mu]} = \frac{\text{cov}[\hat{\mu}_t, y_t | \mu]}{\sigma_y^2}.$$

We show in the online Appendix how to calculate this coefficient as a function of the model parameters.³⁸

Importantly, our numerical solutions indicate that $\hat{\mu}_t$ and y_t are always positively correlated (conditional on μ). This is because a positive innovation in the external state y_t raises (or at least never lowers) the expected value of y_τ for all $\tau \geq t$, and at the same time also raises the expected value of $\hat{\mu}_\tau$ for all $\tau \geq t$ (as illustrated in figures in online Appendix H.1). Since the memory noise has no effect on the evolution of the external state, there are no shocks that move $\hat{\mu}_t$ and y_t in opposite directions, while some (at least the innovation ϵ_{y_t}) move both of them in the same direction. But given that $\beta_{\hat{\mu}|y} > 0$, equation (5.2) implies that $\rho_h^s > \rho$; that is, our model implies overreaction of the kind exhibited by the forecasts of the subjects of Afrouzi et al.

Equation (5.2) also implies that for fixed values of the model parameters other than ρ , the overreaction measure $\rho_h^s - \rho$ converges to zero as $\rho \rightarrow 1$, for any forecast horizon h .³⁹ This is also approximately true of the regression coefficients reported by Afrouzi et al. (2023, Figures 2B and 3). Indeed, these authors stress the finding that in their data, the discrepancy $\rho_h^s - \rho$ is much larger when ρ is relatively small. As will be seen, this is also true in numerical solutions of our model.

Our model can be parameterized so that it simultaneously fits the experimental evidence for the different horizons for which forecasts are solicited in the experiment of Afrouzi et al. We find parameters that fit the regression coefficient estimated from the experimental data. In particular, we target those from one-period forecasts (i.e., ρ_1^s). As discussed in detail in online Appendix H.3, we find pairs of $(\bar{\lambda}, K)$ that minimize the sample MSE in matching the estimated ρ_1^s for six different values of ρ . The targeted moments do not allow separate identification of $\bar{\lambda}$ and K , so we find every pair that is equally good at fitting the moments.

Figure 7 plots the model-predicted moments under our parameterization for both targeted and untargeted moments. The squares show the values of the estimated ρ_h^s against the actual ρ from the experiment of Afrouzi et al., and the solid line corresponds to the model prediction. We find pairs of $(\bar{\lambda}, K)$ that best fit the data points in the top left panel and average the values of ρ_h^s predicted by all such pairs. The other panels show the model fit in generating the extent of overreactions observed in longer horizon forecasts (for $h = 2, 5$, and 10). These are untargeted moments, which speak to the model's ability to quantitatively explain the degree of overreactions that vary with the persistence of the forecasted variable and the forecast horizons.

³⁸ See online Appendix H.2 for details.

³⁹ This prediction depends on $\beta_{\hat{\mu}|y}$ remaining bounded as ρ approaches 1. This is the case in our numerical solutions, both when $\bar{\lambda}$ is held constant as ρ is varied (as in Figure 2) and when θ is held constant as ρ is varied (as in Figure 4).

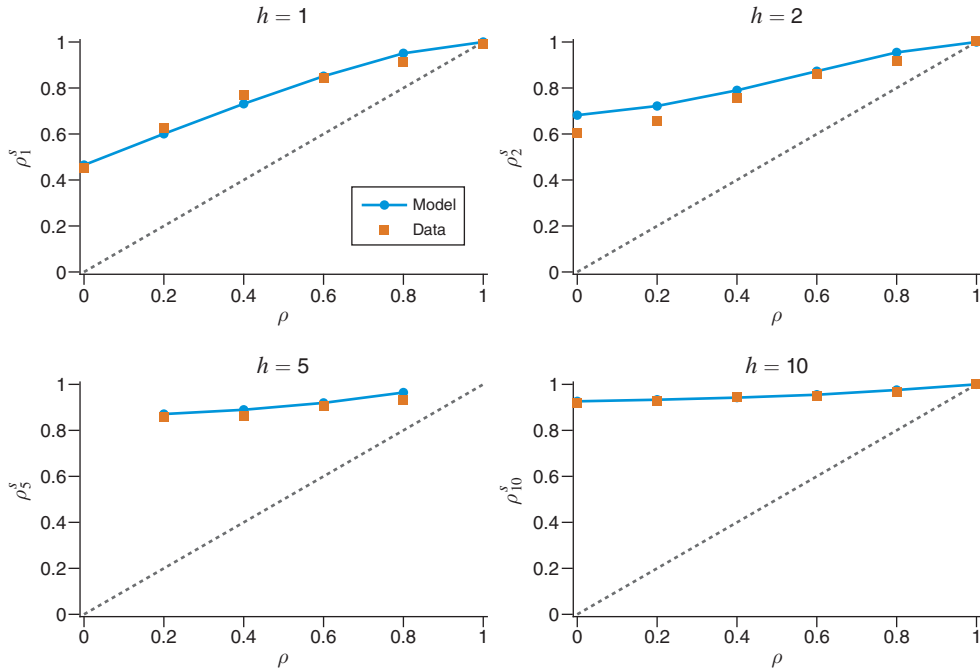


FIGURE 7. TWO REGRESSION COEFFICIENTS

Notes: Comparison of the values for the regression coefficients ρ and ρ_h^s for different values of ρ and h . Model parameters are calibrated to match data points on the top-left panel. The diagonal line indicates the prediction of the rational expectations hypothesis.

There is also evidence of overreaction to news in the forecasts of macroeconomic and financial variables by professional forecasters, as discussed by Bordalo et al. (2020). A satisfactory quantitative account of the predictable forecast errors observed in these forecasts requires an extension of the model presented here, as discussed by Sung (2024). While the more complex model in that paper involves additional information frictions, as addition to allowing for more complex dynamics of the variables that are forecasted, noisy memory of the kind modeled here remains crucial for explaining the observed patterns. And while information frictions of the kind proposed by Coibion and Gorodnichenko (2012, 2015) are also important, Sung finds that quantitative estimates of the size of those frictions are significantly biased by failing to take account of the effects of noisy memory.

VI. Related Models

Here we compare our model to alternative models of belief formation that make at least somewhat similar predictions, most notably with regard to the possibility of overreaction to recent news. We show how our model has important formal similarities to some of these others, and clarify the ways in which it differs from them.

A. Alternative Explanations for Overreaction

We begin by reviewing possible explanations for overreaction to news that do not rely upon imperfect memory. The following subsection then considers alternative models of imperfect memory.

Reactions to News when the Mean Is Understood to Drift.—In our model, overreaction of forecasts to new observations of the variable y_t reflect revisions of the DM's estimate of the mean of the stochastic process $\{y_t\}$, even though the mean μ is assumed to be constant over time; failure of the DM to learn the exact value of μ , even in the long run, depends on imperfect memory. However, there would be a perpetual revision of beliefs about the mean, even with perfect memory (and perfect Bayesian inference) in a world where the mean is (correctly) understood to evolve stochastically over time. In this case, observations farther in the past would be of progressively less relevance to the DM's current estimate of the mean, even with perfect memory.

Such a model can predict forecast dynamics similar (though not identical) to those in our model. As a simple example, suppose that $y_t = \mu_t + \epsilon_t$, where $\epsilon_t \sim N(0, \sigma_y^2)$ represents an i.i.d. deviation from the time-varying mean μ_t ; and suppose that the mean evolves according to an AR(1) process,

$$\mu_t = \phi\mu_{t-1} + \nu_t,$$

where $0 < \phi < 1$ and $\nu_t \sim N(0, (1 - \phi^2)\Omega)$ is another i.i.d. process. Note that this specification implies that the unconditional prior distribution for the mean is given by $N(0, \Omega)$, just as in our model.⁴⁰ Let us consider the evolution of the beliefs of a perfect Bayesian DM in such an environment, who observes y_t with perfect precision each period, starting from prior beliefs about μ_0 (before y_0 is observed) corresponding to the unconditional prior.

The posterior distribution for the value of μ_t , just before the observation of y_t , will be a Gaussian distribution $N(\hat{\mu}_{t|t-1}, \hat{\sigma}_{t|t-1}^2)$. The posterior after observing y_t will be another Gaussian distribution $N(\hat{\mu}_{t|t}, \hat{\sigma}_{t|t}^2)$. The mean and variance of this distribution are given by the same Kalman filter formulas (2.1) and (2.3).⁴¹ In the case of perfect memory, these posterior beliefs about μ_t imply a posterior distribution for the value of μ_{t+1} that is also Gaussian, with mean and variance

$$(6.1) \quad \hat{\mu}_{t+1|t} = \phi \hat{\mu}_{t|t},$$

$$(6.2) \quad \hat{\sigma}_{t+1|t}^2 = \hat{\sigma}_{t|t}^2 + (1 - \phi^2)(\Omega - \hat{\sigma}_{t|t}^2).$$

Comparison of equation (6.2) with the corresponding equation (2.6) for the dynamics of posterior uncertainty in our noisy-memory model reveals that the

⁴⁰The difference is that in the drifting-mean model, we no longer assume that a value of the mean is drawn from this distribution and then remains constant forever after. Our specification in the previous sections can be regarded as the $\phi \rightarrow 1$ limit of this prior.

⁴¹Here we must substitute $\hat{\mu}_{t|t-1}$ for the prior mean $\bar{m}_t$ in equation (2.1), and $\hat{\sigma}_{t|t-1}^2$ for the prior variance Σ_t . Similarly $\hat{\mu}_{t|t}$ and $\hat{\sigma}_{t|t}^2$ correspond to the variables called simply $\hat{\mu}_t$ and $\hat{\sigma}_t^2$ in the previous equations.

degree of uncertainty, after any particular number of observations of y_t , is the same in both models in the case that $\phi^2 = \bar{\lambda}$. The same path for uncertainty about the mean then implies the same path for the Kalman gain γ_t , given by (2.2). Hence the perfect Bayesian model with a stochastic mean is equally capable of explaining why a DM's estimate of the mean should continue to be influenced by recent observations, even after a long sequence of observations; the predictions of the two models about this are identical, if the parameter ϕ is chosen appropriately.

However, this does not mean that the two models are observationally equivalent. Equations (6.1) and (6.2) together with the Kalman filter equation (2.1) imply that after a sufficiently long sequence of observations, the Bayesian estimate of the mean will be given by a solution of the form (2.10), but with the weights $\{\alpha_j\}$ given by

$$\alpha_{j,t} = \phi^j \bar{\gamma} (1 - \bar{\gamma})^j,$$

where $\bar{\gamma}$ is a constant value between 0 and 1, and without any idiosyncratic noise. If we assume that $\phi^2 = \bar{\lambda}$, so that the Kalman gains implied by both models are the same, the weights on past observations $\{y_\tau\}$ will equal

$$\alpha_{j,t} = \bar{\lambda}^{j/2} \gamma_\infty (1 - \gamma_\infty)^j.$$

The weights decay exponentially, like the weights implied by the noisy-memory model; but they do not decay at the same rate. (The weights decay more rapidly as j increases in the case of the noisy-memory model; hence the weights are smaller for all $j \geq 1$ in that case.)

Another difference between the two models is that the perfect Bayesian model implies that there should be a tight relationship between the degree of persistence of the series $\{\mu_t\}$ —and hence the autocorrelation of the observed series $\{y_t\}$ —and the coefficients (such as the Kalman gain γ_t) that describe the dynamics of beliefs about the mean. In the noisy-memory model, the coefficient $\bar{\lambda}$ that determines the size of the Kalman gain and the intrinsic persistence of the belief state can be specified independently of the time series properties of the process $\{y_t\}$. This flexibility is important for accounting for observed beliefs. Bayesian models of subjective forecasts often have to posit a DM with an apparent prior belief that an unknown state fluctuates more than is actually the case.⁴² The noisy-memory model can account for such findings without having to suppose that people fail to learn the correct statistics of their environment. The example just presented shows that noisy memory ($\bar{\lambda} < 1$) can result in belief dynamics similar to those of a Bayesian model in which the DM's prior assumes that the mean μ_t is more stochastic than it really is (the prior assumes that $\phi < 1$ when actually μ never changes).

Finally, the perfect Bayesian model implies that the DM's estimate $\hat{\mu}_{t|t}$ (and hence forecasts of future y_t) at any time will be a deterministic function of the sequence of values $(y_0, \dots, y_t)$ that have been observed. It follows from this that all forecasters who observe the same series should have identical forecasts, and that the variation

⁴² See, for example, Yu and Cohen (2008).

over time in their forecasts can be fully accounted for by the variation in the values that have been observed. The noisy-memory model instead implies that each DM's beliefs and forecasts are affected by memory noise, as shown in (2.10); this implies both that forecasts are not perfectly predictable from the past history of the series being forecasted, and that they should differ across forecasters. This is an attractive feature of the noisy-memory model, since observed forecasts have both of these properties.⁴³

Incomplete Learning Owing to a Small Sample.—Another reason for overreaction of forecasts to new observations (relative to the RE prediction, assuming knowledge of the true mean) could be that the DM has not observed a long enough history of the series $\{y_t\}$ to have learned the mean with sufficient precision, even though the DM's forecasts are optimal conditional on what they have seen. An obvious question about this proposal is why the DM should not have access to a long enough history (under the assumption of perfect memory). Presumably the reason must be that the process $\{y_t\}$ has not been generated according to the current law of motion since time immemorial, and this is evident to the DM, so that only a relatively short series of observations is considered relevant for the current forecast. Overreaction could continue to be observed over time, under this justification, only if one supposes that the world continues to change from time to time, so that the relevant sample is never too long.

This explanation would then be closely related to the hypothesis of parameter drift discussed in the previous subsection, and subject to similar criticisms. In addition, while such a model would allow for predictable forecast errors from the standpoint of an observer who knows the true value of μ at each point in time, there would not be any errors that are forecastable on the basis of variables actually observed by the DM (either currently or in the past). To the extent that forecast errors are found to be correlated with variables known to have been observed by the forecaster (as for example in the study of Bordalo et al. 2020), this finding is inconsistent with the hypothesis of correct conditioning on a sample that is never too large.⁴⁴

Constant-Gain Learning.—One also obtains a prediction of perpetual learning, and hence continued overreaction to news even after an arbitrarily long sequence of observations, in a model where the DM is assumed to estimate the value of the parameter μ using a “constant-gain” variant of least-squares learning (Evans and Honkapohja 2001, sec. 7.4). Constant-gain (CG) algorithms effectively put an exponentially decreasing weight on observations farther in the past; for example, an unknown mean is estimated by a linear estimator of the form

$$\hat{\mu}_t = \sum_{j=0}^{\infty} \gamma (1 - \gamma)^j y_{t-j},$$

where $0 < \gamma < 1$ is the constant “gain factor. This is similar to the kind of estimate of the unknown mean implied by the perfect Bayesian model in the case of

⁴³ See Sung (2024) for discussion of the difference between individual professional forecasters' forecasts and the consensus forecast, in the case of a variety of macroeconomic variables.

⁴⁴ The finding of Bordalo et al. (2024) is instead consistent with the model of noisy memory proposed here, as discussed in Sung (2024).

a drifting mean.⁴⁵ Moreover, in the CG algorithm, the value of γ can be specified independently of the dynamics of the process $\{y_t\}$ that is forecasted.⁴⁶

To the extent that a model of CG learning is considered to be empirically realistic, however, a question arises as to what determines the value of the gain parameter. In the adaptive control literature, such algorithms are proposed as a way of dealing with drift in the values of parameters to be estimated; the appropriate value of the gain parameter should thus depend on one's prior regarding the degree of volatility of the parameters to be estimated, as in our discussion above of Bayesian inference when the mean drifts. But once again, the gain parameters that are found to best fit expectations data do not seem to correspond to ones that would be optimal given the degree of structural change in the forecasted time series.⁴⁷ Alternatively, authors such as Malmendier and Nagel (2016) propose that aggregate dynamics similar to those predicted by a model of CG learning can result from aggregation of the decisions of people of different ages, who each form beliefs on the basis of their personal experience (and hence on the basis of samples extending different distances into the past).⁴⁸ But also under this explanation for CG learning, the predicted gain parameter should depend on other features of the model, that may not justify a gain parameter as large as the one required to explain the observed degree of overreaction to news.⁴⁹ Our model provides an alternative foundation for belief dynamics similar to those implied by a CG algorithm, in which a substantial gain parameter can exist even when the value of the mean remains constant (or nearly constant) over long periods of time, and even when forecasters have long personal histories of observations.

Forecasts Based on an Incorrect Model.—A longstanding explanation for systematic overreaction to news is the hypothesis that people form their forecasts on the basis of an incorrect statistical model—for example, under an assumption that the fluctuations in $\{y_t\}$ are more persistent than is actually the case. Explanations of this kind have continued to be prominent in the recent literature (e.g., Angeletos et al. 2021), but they raise the question: why should people persist in misestimating the dynamics?

Fuster, Laibson, and Mendel (2010) and Fuster, Hébert, and Laibson (2011) offer one answer: people's forecasts are optimal, given their estimated model of the dynamics, and their estimated model is the one that best fits the autocorrelation function of the actual series, within some parametric family of possible models (that need not include the true data generating process). Their hypothesis of “natural expectations”

⁴⁵Note that if we consider a limiting case in which $\phi \rightarrow 1$ while $(1 - \phi^2)\Omega \rightarrow \sigma_v^2 > 0$, then the dynamics of the mean estimate implied by the perfect Bayesian model are exactly those of a constant-gain mean estimate with a gain factor of $\bar{\gamma}$.

⁴⁶In empirical applications (e.g., Milani 2007, 2014; Slobodyan and Wouters 2012), the gain parameter and the parameters specifying the persistence of the exogenous states are treated as independent free parameters to be estimated.

⁴⁷See, e.g., Branch and Evans (2006) and Berardi and Galimberti (2017).

⁴⁸For additional examples, see Nakov and Nuño (2015), Schraeder (2016), Collin-Dufresne, Johannes, and Lochstoer (2017), Ehling, Graniero, and Heyerdahl-Larsen (2018), and Malmendier, Pouzo, and Vanasco (2020).

⁴⁹Thus, Malmendier, Pouzo, and Vanasco (2020) posit an exponentially decaying influence of earlier experiences on a given DM's expectations, even among the events that have occurred during one's lifetime, rather than relying upon demographics alone to account for the rate at which past events cease to influence current market pricing.

assumes that the class of statistical models considered is that of all possible $AR(k)$ models, for some fixed bound on k .⁵⁰ The authors argue that actual time series often involve long-horizon dependencies, and show that in this case (say, an $AR(40)$ process forecasted by people who consider models with no more than 10 lags), long-horizon forecasts using the best fitting $AR(k)$ model can significantly overreact to recent trends in the data.

This proposal, however, remains subject to several objections. Why should the restriction to models of the data with a fixed upper bound on k be maintained, even when the available sequence of observations with which to estimate the model becomes unboundedly long? Moreover, even if one grants that a constraint on model complexity requires that no more than some finite number of explanatory variables be stored and used as a basis for forecasts, why must the possible explanatory variables correspond only to the last k observations of the series? In the kind of example in which Fuster et al. argue that their proposal predicts overreaction, more accurate long-horizon forecasts would be possible if the forecast were conditioned on a long moving average of observations, rather than only recent observations; yet tracking a small number of moving averages would seem no more complex than always having access to the last k observations. And above all, the Fuster et al. explanation implies that overreaction should only be observed in the case of variables that are not well described by an $AR(k)$ process of low enough order. Yet as discussed above, Afrouzi et al. (2023) find significant overreaction in an experiment in which the true data generating process is an $AR(1)$ process; and in fact, they find the most severe degree of overreaction when the process to be forecasted is white noise.

Like the hypothesis of “natural expectations,” our model assumes that forecasts are optimal, among those forecasting rules in which the forecast is based on only a limited summary of past history; but the way in which we model the limit on the complexity of possible representations of past data is different. Our approach does not impose any a priori restriction on either the dimensionality of the memory state or the number of past observations that can be (imperfectly) represented by the memory state. And the form of complexity limit that we assume has the advantage of implying forecasting bias (and more specifically overreaction) even when the true dynamics are very simple. Indeed, even when the true dynamics are white noise (and are recognized by the DM to be white noise).

B. *Alternative Models of Imprecise Memory*

We are also not the only authors to have proposed that expectational biases may result from forecasts being based on imperfect memory of past observations. Here we briefly discuss similarities and differences of alternative proposals with our own approach.

Models of Quasi-Bayesian Belief Updating.—Nagel and Xu (2022) propose that a variety of asset pricing anomalies can be explained by the biases in expectations regarding future asset returns implied by a particular type of departure from perfect

⁵⁰More general versions of this hypothesis are considered in the more recent work of Molavi (2023).

Bayesian inference from observed past returns, which they call a model of “fading memory.” As in this paper, they consider a situation in which a DM (an investor) must infer the mean μ of a process $\{y_t\}$, based on past observations of this process; and (as in the simple case analyzed in Section II) they assume that the process is i.i.d. (and known to be), and that the only unknown parameter of the distribution is μ . Given a prior $p(\mu)$ over possible values of μ , and a likelihood $p(y|\mu)$ for the observation of y_t in any period conditional on the unknown mean, the Bayesian posterior distribution conditional on a finite sequence of observations $\mathbf{y} = (y_{t_0}, \dots, y_{t-2}, y_{t-1}, y_t)$ is given by

$$p(\mu|\mathbf{y}) \sim p(\mu) \prod_{j=0}^{t-t_0} p(y_{t-j}|\mu).$$

The Nagel and Xu (2022) model of “fading memory” instead assumes a subjective posterior of the form

$$(6.3) \quad p(\mu|\mathbf{y}) \sim p(\mu) \prod_{j=0}^{t-t_0} p(y_{t-j}|\mu)^{(1-\nu)^j},$$

for some small quantity $\nu > 0$, which indicates the rate at which memory of past observations “fades.” (Note that their model reduces to perfect Bayesian inference in the limiting case in which $\nu = 0$.)

The Nagel and Xu (2022) model, like ours, is one in which there is perpetual learning: in the limit as $t_0 \rightarrow -\infty$, the posterior distribution (6.3) remains nondegenerate, despite being based on a sample of infinite length. As in our case, the reason is that past observations have a progressively weaker influence on the posterior, the farther they are in the past, and more specifically the influence decreases as an exponential function of the elapsed time. Also as in our case, the Nagel-Xu model implies that one should observe “recency effects.” Another important similarity between their approach and ours is that Nagel and Xu model the DM’s complete posterior at each point in time, not just the DM’s point estimate of μ ; and like us, they tie the rate of decay of past information to cognitive limitations, rather than the rate at which the environment is objectively likely to have changed.

Our model differs from that of Nagel and Xu (2022), however, in offering an explicit representation of the imprecise information contained in memory, and then deriving the DM’s subjective posterior from (correct) Bayesian conditioning on this imprecise record, rather than directly assuming a particular modification of the Bayesian expression for the posterior beliefs. This is not simply a matter of having failed to provide intermediate steps in the derivation; the subjective beliefs assumed by Nagel and Xu are not correct conditional beliefs, if one were to condition on the information about past observations reflected in the assumed beliefs (and therefore revealed by the DM’s cognitive state, since the subjective posterior must be some function of the cognitive state). Our model also differs from theirs in that it implies that individuals’ beliefs involve idiosyncratic cognitive noise; thus, our model, unlike that of Nagel and Xu, predicts that investors should have heterogeneous beliefs even if they observe identical information. (This difference is relevant for applications to financial economics, since our model of heterogeneous beliefs on the part of individual investors provides a motive for trading, even when all information about asset fundamentals is public.) In these respects, the predictions of our

model are not quantitatively identical to those of the model of Nagel and Xu, despite many similarities.

Prat-Carrabin et al. (2021) derive a quasi-Bayesian posterior very similar to the one postulated by Nagel and Xu from a hypothesis of “costly Bayesian inference,” in which belief updating after each new piece of evidence arrives is distorted (relative to exact Bayesian updating) so as to reduce the precision of the resulting belief state.⁵¹ This hypothesis is even more closely related to the one that we propose here, insofar as the sensitivity of beliefs to past observations decreases over time as a consequence of the cost of storing a more precise record of the DM’s past cognitive state. The model of Prat-Carrabin et al. (2021) differs from ours in identifying the imprecise memory state with the DM’s (distorted) posterior beliefs given the sequence of observations to that point; instead, we distinguish between the DM’s cognitive state (which includes the memory state m_t) and the probability beliefs that would optimally be inferred from such a state. Thus, while there are again many similarities between the predictions of their model and ours, the predictions are not identical.

Alternative Models of Noisy Memory.—Neligh (2024) proposes a model of decaying memory that is conceptually closer to our own in that, as in this paper, it is assumed that memory can be retrieved only with noise, and the judgments that are made are optimal (consistent with correct Bayesian inference) subject to being based on the noisy memory state. The difference with our model is in how the memory state and the cost of retrieving a more precise memory are modeled.

Neligh (2024) assumes an “episodic” memory, in which there is an independent noisy record of each of the past observations y_τ for $0 \leq \tau \leq t - 1$; the element of the memory vector corresponding to the observation at time τ is equal to the value of y_τ plus a mean-zero Gaussian noise term, distributed independently of the value of y_τ , and with a variance that depends on the amount of elapsed time. This is a special case of the kind of noisy memory that our framework allows for, but is not the form of memory that is found to be optimal for the decision problem considered in this paper. In addition to imposing the constraint that memory must take this form, Neligh (2024) endogenizes the precision of memory in only one respect: the precision with which observation y_τ is initially encoded at date τ is optimized (subject to a cost of greater encoding precision), but given the choice of an initial encoding precision, the precision of the memory that can be retrieved after a time delay is exogenously determined by the amount of time that has elapsed. Our model instead allows the precision with which memory is maintained over time to be endogenously varied.⁵²

An important similarity between Neligh’s model and ours is that in both models, observations more distant in the past are retrieved with greater noise, because of the way in which noise is cumulatively added as the memory state is maintained over time. This means that both models predict recency effects.⁵³ There would remain,

⁵¹ Prat-Carrabin, Meyniel, and Azeredo da Silveira (2024) fits the model to an experimental dataset.

⁵² In addition to considering a different class of possible memory structures, Neligh (2024) addresses largely distinct questions from those analyzed here.

⁵³ It would be possible to specify the rate of increase of memory noise with the passage of time in Neligh’s model in such a way as to make the distribution of $E[\mu|m_t]$ conditional on the sequence of past observations—and hence the conditional distribution of all of the DM’s forecasts, in the decision problem considered here—the same as the one predicted by our model.

however, two important differences between Neligh's model and ours. One is that our model derives its predictions from less special assumptions and involves fewer free parameters; thus, in the case that both models were equally consistent with empirical observations like those of Afrouzi et al. (2023), our model would provide a more parsimonious explanation. And second, Neligh's model implies a much higher dimensional memory state than does ours. In the case of the decision problem considered in this paper, this makes no difference, as forecasts depend on memory only through a single scalar summary statistic; but the predictions of the two models would likely be different in the case of more complex decisions.

Like us, Afrouzi et al. (2023) propose to explain the biases in their experimental subjects' forecasts using a model of endogenously imprecise memory. The nature of the imprecise representation that is used for the decision is optimized subject to a cost of precision, which as in our model is based on mutual information (for them, the mutual information between the complete contents of memory and the imprecise representation). As in our model, the information cost implies that an accurate estimate of the value of μ cannot be made on the basis of memory, even after a very large number of observations. Hence subjects' forecasts (assumed as in our model to be optimal subject to having to be conditioned on an imprecise cognitive state) are based on a precise observation of the current y_t together with an imprecise estimate of μ deriving from an imprecise summary of past observations. This results, as in our model, in a prediction of overreaction to the most recent observation (that can be observed with greater precision than any past observations are recalled); and the predicted degree of overreaction is greatest in the case of variables with low persistence (since in this case optimal forecasts are largely determined by the optimal estimate of μ).

Despite these similarities in the predictions of the two models, there is an important difference between the model of noisy memory in Afrouzi et al. (2023) and our own. In their model, all past observations are stored in memory with perfect precision; imprecision enters only when an imperfect representation of the contents of memory is retrieved in order to inform a decision. Thus, information retrieved from memory (at a cost) for use in a decision at time t has no consequences for the information that will be available later. This makes the problem of optimal selection of the information to be retrieved at any time t a (relatively simple) static problem in their model, whereas it is a dynamic problem in the model proposed here, since in our model, information not remembered at time t cannot (at any cost) be retrieved in any later period.

This difference in their model implies that all past observations are accessible with equal precision when a forecast needs to be made; hence the optimal noisy representation implies much less "decay" of old observations than in our model. As a simple example, consider the case in which y_t is i.i.d. Then the contents of memory will be distributed independently of the current observation y_t , and the equally weighted sample mean of the observations $\{y_\tau\}$ for $0 \leq \tau < t$ will be a sufficient statistic for the information about the mean μ that is contained in the previous observations; hence the optimal representation will be a noisy read-out of this sample mean. It follows that any past observation y_τ (for $\tau < t$) should have exactly the same effect on forecasts at time t as any other: there will be no "recency effect" at all, except for the fact that the observation y_t will have a larger

effect than any of the observations at dates $\tau < t$. Thus, the model of Afrouzi et al. provides no explanation for the kind of recency effects that have frequently been documented in the experimental literature (e.g., Hogarth and Einhorn 1992), as well as in macroeconomic and financial contexts by authors such as Malmendier, Pouzo, and Vanasco (2020).

VII. Conclusion

We have shown that it is possible to characterize the optimal structure of memory, for a class of linear-quadratic-Gaussian forecasting problems, when the cost of a more precise memory is increasing in Shannon's mutual information, and when we assume that the joint distribution of past cognitive states and the memory state is of a multivariate Gaussian form, but with no a priori restriction on the dimension of the memory state or the dimensions of past experience that may be more or less precisely recalled. Strikingly, we find that for the class of problems that we consider, the optimal memory structure is necessarily at most one-dimensional. This means that what can be recalled at any time about past observations is simply a noisy recollection of a single summary statistic for past experience. We show how the model parameters determine the law of motion for that summary statistic, and hence what single dimension of past experience will be (imprecisely) available as an input to the DM's forecasts.

Among the implications of our model, two seem of particular general interest. First, while our formalism allows for the possibility of an independent noisy record of each past observation (as assumed for example in the model of Neligh 2024), this is not optimal; instead, the optimal memory structure is one in which only a particular weighted average of past observations can be recalled with noise. And second, this weighted average places much larger weights on recent observations than on ones at earlier dates, even though observations at all dates are equally relevant to inference about the value of the parameter μ , which matters for the DM's decisions. Thus, our model provides an explanation for "recency bias" in the influence of past observations on current decisions, unlike the model of endogenous memory precision proposed by Afrouzi et al. (2023).

In the applications sketched above, we have focused on biases that have been observed in people's stated expectations. But we suspect that the expectational biases implied by our model can help to explain puzzling aspects of market outcomes as well. For example, Bordalo et al. (2024) argue that a number of well-known puzzles about the behavior of the aggregate stock market are in fact all consistent with a simple dividend discount model of stock prices, under the hypothesis that market expectations regarding firms' future earnings differ systematically from rational expectations in a particular way, in a way that is also indicated by the biases observed in survey expectations of earnings.

They further show that a particular sort of bias in market expectations is needed in order to explain both the biases in survey expectations and the asset pricing anomalies, one very much like the kind of forecast bias predicted by our model. The most important difference with the reduced-form specification of expectational bias proposed by Bordalo et al. (2024) is that in their model, there are arbitrary random variations in the "market expectations" that determine the value of the stock market; our

model instead implies the existence of idiosyncratic random variation in the beliefs of an individual forecaster, but one might expect that these idiosyncratic variations should cancel out in their effects on the market price. It is possible that a satisfactory model of asset pricing will require us to suppose that some individual traders are large enough for their idiosyncratic beliefs to have a nonnegligible effect on aggregate outcomes, as in the model of Gabaix et al. (2006). We leave the development of a complete model of asset prices for future work; but it seems likely that imperfect memory of the kind modeled here will be a necessary element in such a model.

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